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Contract Checking as Effects

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Anh Khoa Nguyen

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ABSTRACT

Contract Checking as Effects

Anh Khoa Nguyen

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Acknowledgements

Text for acknowledgments goes here. You can thank your advisors, peers, and family.

Preface

This is the preface (optional).

Dedication

This is the dedication (optional).

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CHAPTER 1

Introduction

Software contracts, or *behavioral contracts*, allow programmers to attach assertions to values, and functions. These assertions are runtime guarantees, on violation, an error is thrown, blaming the violating party. Contracts are most useful when placed on a function, enforcing the function’s input and output values. While first-order functional contracts are useful, they are too simple. Higher-order functional contracts allows programmers to enforce contracts for higher-order functions. Dependent contracts are functional contracts allowing usage of the function argument during post-condition contract checking.

Recently, algebraic effect handlers has gained a large attraction, and is adopted in OCaml since version 5.0. Effect handlers open a new way to identify effects and handling them. Handlers are defined to handle one or many effects. A handler for an effect can stop the program or continue with some result. Handlers do not have state by default, but it can be implemented by applying the new states to the continuation closure, and let the underlying computation receives a state argument.

Combining the contract system and effect handlers is a new approach. Effectful Contract [11] demonstrates how both systems can be combined, providing contracts for effectful code. Contracts in their system remain a language feature. In this dissertation, instead of combining the effects and contracts, we ask ourselves whether contract checking itself can become an effect, and reuse the capabilities of handlers.

Contract checking can be encoded as effects, and delegate them to handlers can unify all contracts in the same context, while separated from main code.

1.1 Contributions

This dissertation provides concrete evidence to the thesis claim. We model the two languages **Contracts** representing software contracts, and **Effects** representing effect handlers. Then a formal model for a compiler between **Contracts** and **Effects** is presented. This compiler is proven correct, and effect safe. Moreover, we provide an implementation of such compiler by providing a preprocessor to convert OCaml code annotated with contracts into run-able OCaml code based off this compiler model. Lastly, we provide a Redex PLT model to test our theorems.

Structure ...

1.2 Base Language

We define Base language and its syntax in Figure 1. The language extends the PCF language with tuples and mutable cells. This will be the base language for both **Contracts** and **Effects**.

Base

$\tau ::= o$	$n ::= 0 \mid -1 \mid 1 \mid \dots$
$\mid \tau \rightarrow \tau$	
$\mid \langle \tau, \tau \rangle$	$b ::= \text{true} \mid \text{false}$
$\mid \text{ref}(\tau)$	
$\mid \text{unit}$	$v ::= \text{unit} \mid \text{loc}$
	$\mid n \mid b \mid \langle v, v \rangle \mid x \mid \lambda x. e$
$o ::= \text{num}$	$E ::= \square$
$\mid \text{bool}$	$\mid E + e \mid v + E \mid E - e \mid v - E$
	$\mid E \wedge e \mid v \wedge E \mid E \vee e \mid v \vee E$
$e ::= n \mid b \mid \langle e, e \rangle \mid x$	$\mid E e \mid v E \mid \text{if } E \text{ then } e \text{ else } e \mid \text{zero?}(E)$
$\mid \lambda x : \tau. e \mid \mu x : \tau. e$	$\mid \langle E, e \rangle \mid \langle v, E \rangle, \text{inj.l}(E) \mid \text{inj.r}(E)$
$\mid e + e \mid e - e \mid e \wedge e \mid e \vee e$	$\mid \text{new}(E) \mid \text{get}(E) \mid \text{set}(E, e) \mid \text{set}(v, E)$
$\mid e e \mid \text{if } e \text{ then } e \text{ else } e$	
$\mid \text{zero?}(e)$	
$\mid \text{inj.l}(e) \mid \text{inj.r}(e)$	
$\mid \text{new}(e) \mid \text{get}(e) \mid \text{set}(e, e)$	

Figure 1: Base Language

$$\begin{array}{c}
\frac{}{\Gamma \mid \Sigma \vdash n : \text{num}} \text{T-Num} \qquad \frac{}{\Gamma \mid \Sigma \vdash b : \text{bool}} \text{T-Bool} \\
\\
\frac{\Gamma \mid \Sigma \vdash e_1 : \tau_1 \quad \Gamma \mid \Sigma \vdash e_2 : \tau_2}{\Gamma \mid \Sigma \vdash \langle e_1, e_2 \rangle : \langle \tau_1, \tau_2 \rangle} \text{T-Tuple} \qquad \frac{x : \tau \in \Gamma}{\Gamma \mid \Sigma \vdash x : \tau} \text{T-Var} \\
\\
\frac{\Gamma, x : \tau_1 \mid \Sigma \vdash e : \tau_2}{\Gamma \mid \Sigma \vdash \lambda x : \tau_1. e : \tau_1 \rightarrow \tau_2} \text{T-Lambda} \qquad \frac{\Gamma, x : \tau \mid \Sigma \vdash e : \tau}{\Gamma \mid \Sigma \vdash \mu x : \tau. e : \tau} \text{T-Fix} \\
\\
\frac{\Gamma \mid \Sigma \vdash e_1 : \tau_1 \rightarrow \tau_2 \quad \Gamma \mid \Sigma \vdash e_2 : \tau_1}{\Gamma \mid \Sigma \vdash e_1 \ e_2 : e_2} \text{T-App} \\
\\
\frac{\Gamma \mid \Sigma \vdash e_1 : \text{bool} \quad \Gamma \mid \Sigma \vdash e_2 : \tau \quad \Gamma \mid \Sigma \vdash e_3 : \tau}{\Gamma \mid \Sigma \vdash \text{if } e_1 \text{ then } e_2 \text{ else } e_3 : \tau} \text{T-If} \\
\\
\frac{\Gamma \mid \Sigma \vdash e : \text{num}}{\Gamma \mid \Sigma \vdash \text{zero?}(e) : \text{bool}} \text{T-Zero} \qquad \frac{\Gamma \mid \Sigma \vdash e : \langle \tau_1, \tau_2 \rangle}{\Gamma \mid \Sigma \vdash \text{inj.l}(e) : \tau_1} \text{T-Inj-Left} \\
\\
\frac{\Gamma \mid \Sigma \vdash e : \langle \tau_1, \tau_2 \rangle}{\Gamma \mid \Sigma \vdash \text{inj.r}(e) : \tau_2} \text{T-Inj-Right} \qquad \frac{\Gamma \mid \Sigma \vdash e : \tau}{\Gamma \mid \Sigma \vdash \text{new}(e) : \text{ref}(\tau)} \text{T-New-Cell} \\
\\
\frac{\Gamma \mid \Sigma \vdash e : \text{ref}(\tau)}{\Gamma \mid \Sigma \vdash \text{get}(e) : \tau} \text{T-Get-Cell} \qquad \frac{\Gamma \mid \Sigma \vdash e_1 : \text{ref}(\tau) \quad \Gamma \mid \Sigma \vdash e_2 : \tau}{\Gamma \mid \Sigma \vdash \text{set}(e_1, e_2) : \text{ref}(\tau)} \text{T-Get-Cell}
\end{array}$$

Figure 2: Type Rules for Base

$\lambda x.e \ v$	$\rightarrow$	$e[x := v]$	S-App
$\mu x.e$	$\rightarrow$	$e[x := \mu x.e]$	SFix
if true then e_1 else e_2	$\rightarrow$	e_1	S-If-True
if false then e_1 else e_2	$\rightarrow$	e_2	S-If-False
$\text{inj.l}(\langle v_1, v_2 \rangle)$	$\rightarrow$	v_1	S-Inj-Left
$\text{inj.r}(\langle v_1, v_2 \rangle)$	$\rightarrow$	v_2	S-Inj-Right

$\frac{e_1 \rightarrow e_2}{E[e_1], \sigma \rightarrow E[e_2], \sigma} \text{Step}$			
$\frac{\text{loc} \notin \sigma}{E[\text{new}(v)], \sigma \rightarrow E[\text{loc}], \sigma[\text{loc} \rightarrow v]} \text{S-New-Cell}$		$\frac{v = \sigma(\text{loc})}{E[\text{get}(\text{loc})], \sigma \rightarrow E[v], \sigma} \text{S-Get-Cell}$	
$\frac{\text{loc} \in \sigma}{E[\text{set}(\text{loc}, v)], \sigma \rightarrow E[\text{loc}], \sigma[\text{loc} \rightarrow v]} \text{S-Set-Cell}$			

Figure 3: Reduction Rules for Base

CHAPTER 2

Contracts Language

2.1 Example

Contracts provide a run-time validation of programs, similar to how assertions work. Contracts extend basic assertions to support functions, especially higher-order ones, correctness. Past research has move from pre/post condition placing on functions, into obligations [5], or monitors. These constructions support blame tracking, enabling the developer to know *who*, which function, is responsible for a violation.

Figure 4 illustrates a function protected with a monitor. This function should return a value that is even. When applied with 10, it is checked by the function's *domain contract*. Since the function does not impose any properties for its argument, the contract passes. The body of the function is executed normally, but its result must pass the function's *range contract*. Because 11 is not an even number, the program terminates with a blame.

```

    monjk,l(flat(λx. true) → flat(iseven), λx.x + 1) 10
→   guardjk,l(flat(λx. true) → flat(iseven), λx.x + 1) 10
→   monjk,l(flat(iseven), (λx.x + 1) monjl,k(flat(λx. true), 10))
→   monjk,l(flat(iseven), (λx.x + 1) guardjl,k(λx. true, 10))
→   monjk,l(flat(iseven), (λx.x + 1) checkjl((λx. true) 10, 10))
→   monjk,l(flat(iseven), (λx.x + 1) checkjl(true, 10))
→   monjk,l(flat(iseven), (λx.x + 1) 10)
→*  monjk,l(flat(iseven), 11)
→*  blamejk

```

Figure 4: Contracts in action

2.2 Language Model

We define **Contracts**, in Figure 5, by extending the Base language with contracts. Contracts protect an expression, ensuring that the evaluated value conform to the contracts defined. Contracts are defined for all possible types of value. Base values, including booleans and numbers, are trivial to enforce a contract on them. While functions need to check for both its argument and return result. Not only that, functions are first-class citizen in the language, and we can form higher-order functions. A contract for function may want to “see” the argument when checking the result, such is the idea of dependent contracts. Both contracts higher-order functions, and dependent contracts semantics are discussed in [5].

The language supports contracts for tuples. Their contracts will be applied individually during execution. Mutable cells contracts follow the work of [2], where a special contract type `ref/c( $\kappa$ )` is used to protect the cell’s value.

Contracts extends Base	
$\tau ::= \dots$ $\quad \text{con}(\tau)$	$\kappa ::= \text{flat}(e) \mid \kappa \rightarrow \kappa \mid \kappa \rightarrow_i \lambda x : \tau. \kappa$ $\quad \langle \kappa, \kappa \rangle \mid \text{ref/c}(\kappa)$
$e ::= \dots$ $\quad \text{mon}_j^{k,l}(\kappa, e)$ $\quad \text{check}_j^k(e, v)$ $\quad \text{blame}_p$	$v ::= \dots$ $\quad \text{guard}_j^{k,l}(\kappa, v)$ $E ::= \dots$ $\quad \text{mon}_j^{k,l}(\kappa, E)$ $\quad \text{check}_j^k(E, v)$

Figure 5: **Contracts** language

$$\begin{array}{c}
\frac{\Gamma \vdash e : \tau \rightarrow \text{bool}}{\Gamma \vdash \text{flat}(e) : \text{con}(\tau)} \text{Type-C-Flat} \qquad \frac{\Gamma \vdash \kappa_1 : \text{con}(\tau_1) \quad \Gamma \vdash \kappa_2 : \text{con}(\tau_2)}{\Gamma \vdash \kappa_1 \rightarrow \kappa_2 : \text{con}(\tau_1 \rightarrow \tau_2)} \text{Type-C-Fun} \\
\\
\frac{\Gamma \vdash \kappa_1 : \text{con}(\tau_1) \quad \Gamma \vdash \lambda x : \tau_1. \kappa_2 : \tau_1 \rightarrow \text{con}(\tau_2)}{\Gamma \vdash \kappa_1 \rightarrow_i \lambda x. \kappa_2 : \text{con}(\tau_1 \rightarrow \tau_2)} \text{Type-C-DepFun} \\
\\
\frac{\Gamma \vdash \kappa : \text{con}(\tau)}{\Gamma \vdash \text{ref}/c(\kappa) : \text{con}(\text{ref}(\tau))} \text{Type-C-Ref} \\
\\
\frac{\Gamma \vdash \kappa_1 : \text{con}(\tau_1) \quad \Gamma \vdash \kappa_2 : \text{con}(\tau_2)}{\Gamma \vdash \langle \kappa_1, \kappa_2 \rangle : \text{con}(\langle \tau_1, \tau_2 \rangle)} \text{Type-C-Tuple} \\
\\
\frac{\Gamma \vdash \kappa : \text{con}(\tau) \quad \Gamma \vdash e : \tau}{\Gamma \vdash \text{mon}_j^{k,l}(\kappa, e) : \tau} \text{Type-C-Mon} \qquad \frac{}{\Gamma \vdash \text{blame}_l^k : \tau} \text{Type-C-Blame}
\end{array}$$

Figure 6: Type Rules for **Contracts**

When a contract violation occurs, a blame is thrown with the party, caller or callee. The semantics for correct blaming has been studied under [2] and [3]. Following previous works, We use *indy* semantics for dependent contracts. We use $\text{guard}_j^{k,l}(\kappa, v)$ as the run-time proxy to value v procted under a contract κ . Guard is a value, the semantics adapt to usage of guards, and expand further, sometimes back to monitor to evaluate inner expressions. A guard on a basic value will trigger a $\text{check}_j^k(e \ v, v)$ and returns the value v if it passes the predicate e , or returns a blame_j^k to party k with contract location j . The evaluation rules for **Contracts** is presented in Figure 7.

$\text{check}_j^k(\text{tt}, v)$	$\cdots \rightarrow$	v	Check-True
$\text{check}_j^k(\text{ff}, v)$	$\cdots \rightarrow$	blame_j^k	Check-False
$\text{mon}_j^{k,l}(\text{flat}(e), v)$	$\cdots \rightarrow$	$\text{check}_j^k(e \ v, v)$	Mon-Flat
$\text{mon}_j^{k,l}(\kappa, v)$	$\cdots \rightarrow$	$\text{guard}_j^{k,l}(\kappa, v)$	Mon-Guard
where $\kappa \neq \text{flat}(e)$			
$\text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v_1) \ v_2$	$\cdots \rightarrow$	$\text{mon}_j^{k,l}(\kappa_2, v_1 \ \text{mon}_j^{l,k}(\kappa_1, v_2))$	Guard-Func
$\text{guard}_j^{k,l}(\kappa_1 \rightarrow_i \lambda x. \kappa_2, v_1) \ v_2$	$\cdots \rightarrow$	$\text{mon}_j^{k,l}(\kappa_3, v_1 \ \text{mon}_j^{l,k}(\kappa_1, v_2))$	Guard-Dep
where $\kappa_3 = \{\text{mon}_j^{l,j}(\kappa_1, v_2)/x\} \kappa_2$			
$\text{get}(\text{guard}_j^{k,l}(\kappa, v))$	$\cdots \rightarrow$	$\text{mon}_j^{k,l}(\kappa, \text{get}(v))$	Mon-Get
$\text{set}(\text{guard}_j^{k,l}(\kappa, v_1), v_2)$	$\cdots \rightarrow$	$\text{mon}_j^{k,l}(\text{ref}/c(\kappa), e)$	Mon-Set
where $e = \text{set}(v_1, \text{mon}_j^{l,k}(\kappa, v_2))$			
$\frac{E \neq \square}{E[\text{blame}_j^k], \sigma \rightarrow \text{blame}_j^k, \sigma} \text{Step-C-Blame} \qquad \frac{e_1 \cdots \rightarrow e_2}{E[e_1], \sigma \rightarrow E[e_2], \sigma} \text{Step-C}$			

Figure 7: Reduction Rules for **Contracts**

CHAPTER 3

Effects Language

3.1 Example

In Figure 8, we present a simple example: a program wants to compute $(\mathcal{D} \ 2) + 1$ is registered with a handler h for effect $\mathcal{D}$, basically *duplicating* its input. When an effect is perform, the main program pauses the execution, and push everything later into a continuation $k = \lambda y. \text{handle } h \ (y + 1)$, at (i). Here, the continuation register the handler h again, in case the rest of the program uses $\mathcal{D}$. When the program is a value, the handler removes itself (ii).

$$\begin{aligned}
 h &= \{ \mathcal{D} \rightarrow \lambda v. \lambda k. k \ (v + v) \} \\
 &\text{handle } h \ ((\text{perform } \mathcal{D} \ \text{num } 2) + 1) \\
 \Rightarrow &\text{handle } h \ \text{perform } \mathcal{D} \ \text{num } 2 \\
 \Rightarrow &(\lambda v. \lambda k. k \ (v + v)) \ 2 \ (\lambda y. \text{handle } h \ (y + 1)) & (i) \\
 \Rightarrow^* &(\lambda y. \text{handle } h \ (y + 1)) \ (2 + 2) \\
 \Rightarrow^* &\text{handle } h \ 5 & (ii) \\
 \Rightarrow &5
 \end{aligned}$$

Figure 8: Handler in action

Handlers can encode states. This is a nice feature to provide a common state across the code. We illustrate the example in Figure 9, where a simple state can be managed by two effects *get* $\mathcal{G}$, and *set* $\mathcal{S}$. As can be seen, the continuation now returns a function expecting a state. During handler execution, this state is applied automatically if the handler is installed with initial state. The continuation resumes the program with the new state, made possible because handler is installed with new state.

Without careful setup, the program might enter the stuck state because the handler removes itself when the execution completes, making the result applies with the state. For example, this program `handle h v s  $\Rightarrow^*$  v s` sticks if v is not callable. There are many ways to resolve this problem. The first approach requires the program awareness of the state. The programmer explicitly returns a function taking the result and the state but only returns the result.

$$\text{handle } h \ ((\lambda x. \lambda s. x) \ e) \ s \Rightarrow^* \text{handle } h \ ((\lambda x. \lambda s. x) \ v) \ s \Rightarrow^* (\lambda s. v) \ s \Rightarrow^* v$$

The second approach is to use *parameterized handlers* [8], which puts the state into the handler expression. The third approach is to augment the handler with a special effect `return`, which modifies the value before removing the handler. The example Figure 8 chooses the first approach, as it keeps the handler semantics simple, although a programmer needs to pay more attention when dealing with handlers receiving arguments.

$$\begin{aligned} h &= \{ \mathcal{G} \rightarrow \lambda v. \lambda k. \lambda s. k \ s \ s; \mathcal{S} \rightarrow \lambda v. \lambda k. \lambda s. k \ s \ v \} \\ &(\text{handle } h \ ((\lambda x. \lambda s. x) \ ((\text{perform } \mathcal{S} \ \text{num } 42) + (\text{perform } \mathcal{G} \ \text{num unit}) + \\ &100))) \ 0 \\ \Rightarrow^* &(\lambda v. \lambda k. \lambda s. k \ s \ v) \ 42 \ \lambda y. \text{handle } h \ ((\lambda x. \lambda s. x) \ (y + (\text{perform } \mathcal{G} \ \text{num unit}) + 100)) \ 0 \\ \Rightarrow^* &(\lambda y. \text{handle } h \ ((\lambda x. \lambda s. x) \ (y + (\text{perform } \mathcal{G} \ \text{num unit}) + 100))) \ 0 \ 42 \\ \Rightarrow^* &\text{handle } h \ ((\lambda x. \lambda s. x) \ ((0 + (\text{perform } \mathcal{G} \ \text{num unit}) + 100))) \ 42 \\ \Rightarrow^* &\text{handle } h \ ((\lambda x. \lambda s. x) \ 142) \ 42 \\ \Rightarrow^* &(\lambda s. 142) \ 42 \\ \Rightarrow^* &142 \end{aligned}$$

Figure 9: Handler with state

3.2 Language Model

Effects, inspired by [16], equips Base with effects and effect handlers. We extend the type system into a kinded type system based on System F, mainly to separate between effect types and normal types. System F also provides polymorphism. The type system

adapts its syntax of functions into $\tau_1 \rightarrow^\varepsilon \tau_2$ denotes an effect ε (may) occurs during the execution of the function.

Following [16], effects are group of “operations”, distinct by its label. An operation $\mathcal{O}$ of effect ε is invoked by applying **perform** $\mathcal{O} \tau$ with an argument, and produces an effect ε . Given an expression e that would perform some effect ε , a handler h providing all definitions for operations of ε must be installed. The calculus provides two ways to install a handler, **handle** $h e$ and **handler** $h (\lambda_.e)$. The former is the base syntax, during evaluation of e , the handler h is used. The other syntax is useful for suspending a computation, and run it only when a handler is applied, providing flexibility such as different handlers.

A handler is a mapping between operation names and their definitions. An operation is a function with the template $\Lambda\alpha.\lambda x : \tau_1.\lambda k : \tau_2.e$. The polymorphic type variable α is used to make operations polymorphic. The argument x is the operation argument. The argument k is the *continuation* or *resumption*, enclosing the rest of the program after the operation invocation. When k is called, the program resumes with the argument provided to k .

Because k is a function, a handler can return k and let the program use continuation anyhow the programmer wants. This is described in [16] as “undesirable in practice” and define a restriction named *scoped resumption*. Our language also follows the same principle and does not allow returning the continuation k (directly or indirectly through closure). However, we do not enforce this and left out for the language implementators.

Effects extends Base

$\tau ::= \dots$	$v ::= \dots$
$\alpha^k \mid c^k \tau \dots \mid \tau \rightarrow^\varepsilon \tau \mid \forall \alpha^k. \tau$	$\Lambda \alpha^k. v$
blab	handler h
	perform $\text{op } \tau$
	error _{p}
$k ::= *$	$p ::= \text{Blame Party}$
$k \rightarrow k$	
eff	$E ::= \dots$
lab	$E[\tau]$
	handle $h E$
$e ::= \dots$	error _{E}
$e[\tau]$	
handle $h e$	$F ::= \square$
error _{e}	$F + e \mid v + F \mid F - e \mid v - F$
$h ::= \{\text{op} \rightarrow f, \dots, \}$	$F \wedge e \mid v \wedge F \mid F \vee e \mid v \vee F$
	$F e \mid v F$
$f ::= \Lambda \alpha. \lambda x : \tau. \lambda k : \tau. e$	if F then e else e
	$F[\tau]$

Listing 1: **Effects** language

A function can use multiple effects, and are represented as rows with equivalence up to reordering. An empty effect is represented as $\langle \rangle$. Effects can be concatenated by $\langle l, \varepsilon \rangle$ where effect row ε is extended with effect l . For ease of writing, we allow writing $\langle l_1, l_2, \dots, l_n \rangle$ as shorthand instead of $\langle l_1, \langle l_2, \dots \langle l_n \rangle \rangle \rangle$.

The language is explicit about the effects. Lambdas are annotated with effects, as well as perform (operation invocation), and handler installation. These effect annotations are used only during type checking, they do not affect the execution.

Typing for an effect language requires some consideration. Following the works in [16], we define the type judgement relationship $\Delta \vdash e : \tau \mid \varepsilon$ between a type context Δ , an expression e and assert it against type τ with some possible effects ε . Variables and base values do not have effects, they can take any effects. For this reason, we define $\Delta \vdash_{\text{val}} v : \tau$. Handlers are typed using $\Delta \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon$, that it handles a *single* effect l , other effects

in ε remains unhandled, and all operations returns τ . All effects signatures are defined in a global set Σ for fast accessing the type of the operations. This set can be built by going through all handlers, and its existence is assumed during the start of type checking. All type rules unique to **Effects** is presented in Figure 10.

$$\begin{array}{c}
\frac{x : \tau \in \Gamma}{\Gamma \vdash_{\text{val}} x : \tau} \text{T-E-Var} \qquad \frac{\Gamma, x : \tau_1 \vdash e : \tau_2 \mid \varepsilon}{\Gamma \vdash_{\text{val}} \lambda x : \tau_1. e : \tau_1 \rightarrow^\varepsilon \tau_2} \text{T-E-App} \\
\\
\frac{\Gamma \vdash_{\text{val}} v : \tau}{\Gamma \vdash v : \tau \mid \varepsilon} \text{T-E-Val} \qquad \frac{k \neq \text{lab} \quad \Gamma \vdash_{\text{val}} v : \tau}{\Gamma \vdash_{\text{val}} \Lambda \alpha^k. v : \forall \alpha^k. \tau} \text{T-E-TAbs} \\
\\
\frac{\Gamma \vdash e_1 : \tau_1 \rightarrow^\varepsilon \tau_2 \mid \varepsilon \quad \Gamma \vdash e_2 : \tau_1 \mid \varepsilon}{\Gamma \vdash e_1 e_2 : \tau \mid \varepsilon} \text{T-E-App} \qquad \frac{\vdash_{\text{wf}} \tau : k \quad \Gamma \vdash e : \forall \alpha^k. \tau_1 \mid \varepsilon}{\Gamma \vdash e[\tau] : \tau_1[\alpha := \tau] \mid \varepsilon} \text{T-E-Tapp} \\
\\
\frac{\text{op} : \forall \alpha. \tau_1 \rightarrow \tau_2 \in \Sigma(l) \quad \alpha \notin \text{ftv}(\Gamma)}{\Gamma \vdash_{\text{val}} \text{perform op } \tau : \tau_1[\alpha := \tau] \rightarrow^{\langle l | \varepsilon \rangle} \tau_2[\alpha := \tau]} \text{T-E-Perform} \\
\\
\frac{\text{op}_i : \forall \alpha. \tau_1 \rightarrow \tau_2 \in \Sigma(l) \quad \alpha \notin \text{ftv}(\varepsilon, \tau) \quad \Gamma \vdash_{\text{val}} f_i : \forall \alpha. \tau_1 \rightarrow^\varepsilon ((\tau_2 \rightarrow^\varepsilon \tau) \rightarrow^\varepsilon \tau)}{\Gamma \vdash_{\text{ops}} \{\text{op}_1 \rightarrow f_1, \dots, \text{op}_n \rightarrow f_n\} : \tau \mid l \mid \varepsilon} \text{T-E-Ops} \\
\\
\frac{\Gamma \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon}{\Gamma \vdash_{\text{val}} \text{handler } h : (\text{unit} \rightarrow^{\langle l | \varepsilon \rangle} \tau) \rightarrow^\varepsilon \tau} \text{T-E-Handler} \\
\\
\frac{\Gamma \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon \quad \Gamma \vdash e : \tau \mid \langle l | \varepsilon \rangle}{\Gamma \vdash \text{handle } h e : \tau \mid \varepsilon} \text{T-E-Handle}
\end{array}$$

Figure 10: Type Rules for **Effects**

Lastly the semantic model of **Effects** is presented in Listing 2. A handler can be installed to a function as a shorthand, where it applies the function with a unit argument, which to be ignored. The reduction for handling an effect calling is encoded as *deep handler*.

Supporting the shallow should be straightforward by not installing the handler again in the continuation k . There is little advantage to support shallow handlers, therefore, only deep handlers are used. While looking for the handler, we make sure we get the innermost handler that can handle the operation.

$$\begin{array}{lll}
(\Lambda \alpha^k. v)[\tau] & \rightarrow & v[\alpha := \tau] \quad \text{Step-E-TApp} \\
\text{handler } h \ v & \rightarrow & \text{handle } h \ (v \ ()) \quad \text{Step-E-Handler} \\
\text{handle } h \ v & \rightarrow & v \quad \text{Step-E-Return} \\
\\
\frac{\text{op} \notin \text{bop}(E) \wedge (\text{op} \rightarrow f) \in h \quad \text{op} : \forall \alpha. \tau_1 \rightarrow \tau_2 \in \Sigma(l) \quad k = \lambda x : \tau_2[\alpha := \tau]. \text{handle } h \ E[x]}{\text{handle } h \ E[\text{perform op } \tau \ v] \rightarrow f[\tau] \ v \ k} & \text{Step-E-Perform} \\
\\
\frac{E \neq \square}{E[\text{error}_p], \sigma \Rightarrow \text{error}_p, \sigma} & \text{Step-E-Error} & \frac{e_1 \rightarrow e_2}{E[e_1], \sigma \Rightarrow E[e_2], \sigma} \text{Step-E} \\
\\
\textbf{Helper functions} \\
\begin{array}{lll}
\text{bop}(\square) & = & \emptyset \\
\text{bop}(E \ e) & = & \text{bop}(E) \\
\vdots & \vdots & \vdots \\
\text{bop}(\text{handle } h \ E) & = & \text{bop}(E) \cup \{\text{op} \mid (\text{op} \rightarrow f) \in h\}
\end{array}
\end{array}$$

Listing 2: Reduction Rules for **Effects**

CHAPTER 4

Formal Compiler

A compiler from **Contracts** to **Effects** is straightforward for expressions inherited from the Base language. The compiler keeps these same expressions, albeit each sub-expression must be compiled *recursively* to ensure no contracts remains after compilation. Remaining expressions are contract-related expressions. Our idea is that a contract check is a side effect, defined as $\mathcal{C}$. This effect ideally receives a tuple $\langle l, \langle e, v \rangle \rangle$ of blame label l , contract predicate e , and to-be-checked value v . Then, a flat contract is simply an effect perform with its correct inputs.

Functional contracts are more complex. Because the effect can only be invoked on a flat contract, the compiler recursively expands the expression, pads the expression into a function when it is expecting a function. During the compilation, new variables are generated to pad the expression into a function. The process is also applicable to dependent contracts. However, the dependent contract needs to update the post contract with a contract for argument. This is trivial but requires a duplication of compilation with a different label. The rules are crafted carefully to prevent syntax of **Effects** into **Contracts** while compiling the dependent contracts. The compiled argument contract is delayed substitution, using a random variable as placeholder, until we have the **Effects** expression which we can safely replace into.

Compilation for tuple contracts are straightforward. Each tuple entries are compiled individually, and reconstructed into a new tuple. Then, each entries are protected, flat contracts are checked immediately during tuple construction; functions are protected and should be checked when used.

4.1 Handler design

After all flat contracts are converted into performing an effect, we need a handler. A simple handler can be installed like below:

$$h_{\text{naive}} = \{\mathcal{C} \rightarrow \lambda\langle l, \langle e, v \rangle \rangle. \lambda k. \text{ if } e \text{ } v \text{ then } k \text{ } v \text{ else } \text{error}_l\}$$

where $\lambda\langle l, \langle e, v \rangle \rangle$ deconstructs the value into tuple pattern

This handler performs the check for the value v , with the predicate/contract e . The continuation k is invoked with the value if the check passes, and blame the label l if the contract is violated. At first glance, this handler is correct. In fact, the handler should produce correct semantic, as long as no dependent contracts are used. When dependent contracts are used, e is now populated with monitored arguments, which after compilation should include **perform** $\mathcal{C}$ τ **arg**. And since the handler is moved into the continuation k , there is no handler at $e \text{ } v$. Take the following program wrapped with the naive handler

h_{naive} .

$$\begin{aligned}
 & \text{mon}_j^{k,l}(\text{flat}(e_1) \rightarrow_i \lambda y. \text{flat}(e_2), f) \text{ } 42 \\
 & \quad \text{where } e_1 = \lambda x. \text{true}, e_2 = \lambda x. x > y, \text{ and } f = \lambda x. x + x \\
 \hookrightarrow & \quad (\lambda x_1. \text{perform } \mathcal{C} \text{ num } \langle k, \langle e_3, f(\text{perform } \mathcal{C} \text{ num } \langle l, \langle e_1, x_1 \rangle \rangle) \rangle \rangle) \text{ } 42 \\
 & \quad \text{where } e_3 = e_2[y := \text{perform } \mathcal{C} \text{ num } \langle l, \langle e_1, x_1 \rangle \rangle] \\
 \hline
 & \text{handler } h_{\text{naive}} \lambda_. (\lambda x_1. \text{perform } \mathcal{C} \text{ num } \langle k, \langle e_3, f(\text{perform } \mathcal{C} \text{ num } \langle l, \langle e_1, x_1 \rangle \rangle) \rangle \rangle) \text{ } 42) \\
 \Rightarrow & \quad \text{handle } h_{\text{naive}} (\lambda x_1. \text{perform } \mathcal{C} \text{ num } \langle k, \langle e_3, f(\text{perform } \mathcal{C} \text{ num } \langle l, \langle e_1, x_1 \rangle \rangle) \rangle \rangle) \text{ } 42) \\
 \Rightarrow & \quad \text{handle } h_{\text{naive}} \text{perform } \mathcal{C} \text{ num } \langle k, \langle e'_3, f(\text{perform } \mathcal{C} \text{ num } \langle l, \langle e_1, 42 \rangle \rangle) \rangle \rangle \\
 & \quad \text{where } e'_3 = e_2[y := \text{perform } \mathcal{C} \text{ num } \langle l, \langle e_1, 42 \rangle \rangle] \\
 \Rightarrow^* & \quad \text{handle } h_{\text{naive}} \text{perform } \mathcal{C} \text{ num } \langle k, \langle e'_3, f \text{ } 42 \rangle \rangle \\
 \Rightarrow^* & \quad \text{handle } h_{\text{naive}} \text{perform } \mathcal{C} \text{ num } \langle k, \langle e'_3, 84 \rangle \rangle \\
 \Rightarrow^* & \quad \text{if } e'_3 \text{ } 84 \text{ then } (\lambda x. \text{handle } h_{\text{naive}} x) \text{ } 84 \text{ else } \text{error}_k \\
 \Rightarrow^* & \quad \text{if } 84 > (\text{perform } \mathcal{C} \text{ num } \langle l, \langle e_1, 42 \rangle \rangle) \text{ then } (\lambda x. \text{handle } h_{\text{naive}} x) \text{ } 84 \text{ else } \text{error}_k \\
 & \text{stuck}
 \end{aligned}$$

To resolve this issue, we define a recursive function `wrap` that will reinstall the handler at e v making sure all effects during contract checking is handled. Contracts related compilation rules are defined in Figure 11.

4.2 Effects annotation

Careful readers may have noticed that the compiler does not annotate effects. Since the only effects that may occur is the contract checking effect $\mathbf{h}_{\mathcal{C}}$, a lazy compiler can annotate this effect to lambdas, handlers, and performs. By the type system of **Effects**, a function annotated with effect without using the effect can still be typed. Our compiler will produce code annotated with $\langle \mathbf{h}_{\mathcal{C}} \rangle$.

4.3 Compiler Type Preservation and Effect Safety

The compiler should produce same type expression. However, this is problematic when effects are introduced. Recall that the type system separates between pure function, and effectful functions. And type judgement separates between non-effectful computations and effectful computations. Depending on the source, compiled program may have the same type with no effects (no monitors), or same type with some effects (monitors during computation), or purely produce a function wrapped by monitors, or effectfully produce a function wrapped by monitors (monitors during computation).

Theorem 4.1. (Compiler Type Preservation)

If $\cdot \vdash e_1 : \tau \rightsquigarrow e_2$ then $\cdot \vdash e_2 : \tau^* \mid \langle \mathbf{h}_{\mathcal{C}} \rangle$

Theorem 4.2. (Compiler Effect Safety with wrap)

If $\cdot \vdash e_1 : \tau \rightsquigarrow e_2$ and $\cdot \vdash e_2 : \tau^* \mid \langle \mathbf{h}_{\mathcal{C}} \rangle$ then $\cdot \vdash \text{wrap}[\tau^*](\lambda_- : \text{unit}.e) : \tau^* \mid \langle \rangle$

Proof. True by type checking. □

4.4 Compiler Correctness

The compiler should produce a program equivalent to the source program. If the source program produces a value, the target program should also produce (the same) value, and with an equivalent storage space.

Theorem 4.3. (Compiler Correctness)

If $\cdot \vdash e : \tau \rightsquigarrow e'$ and $e^* = \text{untyped}(\text{wrap}[\tau](\lambda_{-}.e'))$ and $e \rightarrow^* b$ then $e^* \Rightarrow^* b$.

$$\begin{array}{c}
 \frac{\Gamma \vdash e_1 : \tau \rightarrow \text{bool} \rightsquigarrow e_3 \quad \Gamma \vdash e_2 : \tau \rightsquigarrow e_4}{\Gamma \vdash \text{mon}_j^{k,l}(\text{flat}(e_1), e_2) : \tau \rightsquigarrow \text{perform } \mathcal{C} \tau \langle k, \langle e_3, e_4 \rangle \rangle} \text{Compile-Flat} \\
 \\
 \frac{\Gamma \vdash e_1 : \tau_2 \rightsquigarrow e_2 \quad \Gamma \vdash \lambda f : \tau_1 \rightarrow \tau_2. \lambda x : \tau_1. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) : (\tau_1 \rightarrow \tau_2) \rightarrow (\tau_1 \rightarrow \tau_2) \rightsquigarrow e_3}{\Gamma \vdash \text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, e_1) : \tau_1 \rightarrow \tau_2 \rightsquigarrow e_3 e_2} \text{Compile-Func} \\
 \\
 \frac{f, x \notin \text{dom}(\Gamma) \cup \{y\} \cup \text{fv}(\kappa_1, \kappa_2) \quad \Gamma \vdash e_1 : \tau_1 \rightarrow \tau_2 \rightsquigarrow e_2 \quad e_w = \lambda f : \tau_1 \rightarrow \tau_2. \lambda x : \tau_1. \text{mon}_j^{k,l}(\{ \text{mon}_j^{l,j}(\kappa_1, x)/y \} \kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) \quad \Gamma \vdash e_w : (\tau_1 \rightarrow \tau_2) \rightarrow (\tau_1 \rightarrow \tau_2) \rightsquigarrow e_3}{\Gamma \vdash \text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, e_1) : \tau_1 \rightarrow \tau_2 \rightsquigarrow e_3 e_2} \text{Compile-Dep} \\
 \\
 \frac{x \notin \text{dom}(\Gamma) \quad \Gamma, x : \langle \tau_1, \tau_2 \rangle \vdash \text{mon}_j^{k,l}(\kappa_1, \text{inj.l}(x)) : \tau_1 \rightsquigarrow e_1 \quad \Gamma, x : \langle \tau_1, \tau_2 \rangle \vdash \text{mon}_j^{k,l}(\kappa_2, \text{inj.r}(x)) : \tau_2 \rightsquigarrow e_2 \quad \Gamma \vdash e : \langle \tau_1, \tau_2 \rangle \rightsquigarrow e_3}{\Gamma \vdash \text{mon}_j^{k,l}(\langle \kappa_1, \kappa_2 \rangle, e) : \langle \tau_1, \tau_2 \rangle \rightsquigarrow (\lambda x. \langle e_1, e_2 \rangle) e_3} \text{Compile-Tuple}
 \end{array}$$

$$\text{h}_{\mathcal{C}} : \text{lab} = \{ \mathcal{C} \rightarrow \forall \alpha. \langle \text{blab}, \langle \alpha \rightarrow \text{bool}, \alpha \rangle \rangle \rightarrow \alpha \}$$

$$\text{wrap} = \mu w : \forall \alpha. ((\ () \rightarrow^{\langle \text{h}_{\mathcal{C}} \rangle} \alpha) \rightarrow \alpha). \Lambda \alpha. \lambda f : (\ () \rightarrow^{\langle \text{h}_{\mathcal{C}} \rangle} \alpha).$$

$$\text{handler } \{ \mathcal{C} \rightarrow \lambda \langle l, \langle e, v \rangle \rangle. \lambda k. \text{ if } w[\text{bool}] (\lambda_{-}.e \ v) \text{ then } k \ v \text{ else } \text{error}_l \} f$$

where $\lambda \langle l, \langle e, v \rangle \rangle$ deconstructs the argument as tuple pattern

Figure 11: Compiler Rules

4.5 Compiling Contracts for Mutable Cells

4.5.1 Observation

Contracts uses $\text{guard}_j^{k,l}(\kappa, v)$ as a proxy to mutable cells. Get and set operation go through this guard to enforce contracts at run-time. This proxy makes the action oblivious whether it is a real cell or a guarded value. Consider this simple function that calls $\text{get}(c)$ for any cell.

$$\lambda c. \text{get}(c)$$

This function should work with both cells $\text{new}(v)$ and guards constructed from $\text{mon}_j^{k,l}(\text{ref}/c(e), r)$ (where r a location). **Effects** does not have special reduction rules for mutable cells, therefore, we need a unified interface between cells and protected cells, and rewrite cell operations to use that unified interface.

4.5.2 Unified Interface

Our compiler has supports for functions and tuples, this is useful for us to design such unified interface. A mutable cell must support get and set, we can encode this as a tuple of 2 functions. Contracts on a cell is now converted into contracts placed on these get and set functions. Cell's location is enclosed inside the closure, and should be accessible when invoked. The compiler should first convert all cell operations into this new wrapper. Then convert all $\text{ref}/c(\kappa)$ into tuple contracts for get and set functions.

$$\frac{\Gamma \vdash e : \tau \multimap e_1}{\Gamma \vdash \text{new}(e) : \text{ref}(\tau) \multimap \text{newcell } e_1} \text{Compile-New}$$

$$\frac{\Gamma \vdash e : \text{ref}(\tau) \multimap e_1}{\Gamma \vdash \text{get}(e) : \tau \multimap \text{getcell } e_1} \text{Compile-Get}$$

$$\frac{\Gamma \vdash e_1 : \text{ref}(\tau) \multimap e_3 \quad \Gamma \vdash e_2 : \tau \multimap e_4}{\Gamma \vdash \text{set}(e_1, e_2) : \tau \multimap \text{setcell } e_3 \ e_4} \text{Compile-Get}$$

$$\frac{\begin{array}{c} x \notin \Gamma \\ \Gamma, x : \langle \text{unit} \rightarrow \tau, \tau \rightarrow \text{loc} \rangle \vdash \text{mon}_j^{k,l}(\langle \text{any} \rightarrow \kappa, \kappa \rightarrow \text{any} \rangle, x) : \langle \text{unit} \rightarrow \tau, \tau \rightarrow \text{loc} \rangle \multimap e_1 \\ \Gamma \vdash e : \text{ref}(\tau) \multimap e_2 \quad e_3 = e_1[x := e_2] \end{array}}{\Gamma \vdash \text{mon}_j^{k,l}(\text{ref}/\text{c}(\kappa), e) : \text{ref}(\tau) \multimap e_3} \text{Compile-Mon-Ref}$$

$\text{newcell} = \lambda v. (\lambda l. \langle \lambda_. \text{get}(l), \lambda v. \text{set}(l, v) \rangle) \text{new}(v)$
 $\text{getcell} = \lambda b. \text{inj.l}(b) \text{unit}$
 $\text{setcell} = \lambda b. \lambda v. (\lambda_. b \ (\text{inj.r}(b) \ v))$

Listing 3: Compiler Rules for Mutable Cells

CHAPTER 5

Implementation

We implement the formal compiler model into OCaml, by adding a preprocessor. We utilize the OCaml attribute to add contract annotations to variables. The preprocessor rewrites annotated variables according to the compiler rules defined. Note that the compiler does not place a handler during compilation, but after the compilation. Because the handler must be installed around the whole program, more precisely, around everywhere the contract annotated variables are used. The developer can install the handler manually by invoking a predefined wrapper, or annotate a function as `main` and the preprocessor rewrites to add wrapper automatically.

Annotations are OCaml annotations. Although OCaml annotations support string-like payload, we find it better to use a OCaml type expression as payload to define contracts. Annotations are not code, therefore we cannot support first-order contracts, and use function names as predicates. In Listing 4, we define the contract annotation syntax that developers can use to annotate their OCaml code with contracts.


```

1  (* flat contract *)
2  let [@contract : predicate] x : t = e
3
4  (* function contract
5   * arguments are inside a tuple
6   *)
7  let [@contract : (p1_pred * p2_pred) -> f_pred]
8    f (p1 : t1) (p2 : t2) : t = f_body
9
10 (* dependent function contract
11  * similarly to function contract, annotated with as 'dep
12  *)
13 let [@contract : (p1_pred * p2_pred) -> g_pred as 'dep]
14   g (p1 : t1) (p2 : t2) : t = g_body
15
16 (* tuple contract
17  * using tuple syntax and annotated with as 'tuple
18  *)
19 let [@contract : (t1_pred * t2_pred) as 'tuple]
20   tup : (t1 * t2) = (e, e)

```

Listing 4: OCaml Contract Annotation Example

The preprocessor rewrite using the compiler rules. All flat contracts are compiled into a single effect, formally it is $\mathcal{C}$, implementation wise we define a name `Contract.Check`.

```

1  module Contract = struct
2  type _ Effect.t +=
3    | Check : (string * ('a -> bool) * 'a) -> 'a Effect.t
4  end

```

`wrap` is defined similarly in OCaml, we remove the type annotations for clarity.

```

1  let rec wrap thunk =
2    Effect.Deep.match_with thunk ()
3    {
4      Effect.Deep.retc = (fun x -> x);
5      exnc = (fun e -> raise e);
6      effc = fun eff ->
7        let open Effect.Deep in
8        match eff with

```

```

9      | Flat (label, check, arg) ->
10      Some (fun k ->
11          let ok = wrap (fun () -> check arg) in
12          if ok
13          then continue k arg
14          else discontinue k (Blame label))
15      | _ -> None
16  }

```

5.1 Contract States and Interaction

5.1.1 Global State

Because the entire program is wrapped into a single handler for all effects, we can unify a single state for effects only. Predicates must now receive an extra argument, which is the context. The wrap function is now defined with an extra argument current context, and it returns the result and updated context. At the callsite, it is provided with the initial value. States are provided to the handler, therefore the handler provide two effects to get and set the state.

```

1  type state = int
2  let state_init = 0
3
4  let rec wrap_inner : type r. (unit -> r) -> state -> r * state =
5      fun thunk init ->
6          match_with thunk ()
7      {
8          retc = (fun x state -> (x, state));
9          exnc = (fun e _state -> raise e);
10         effc = fun (type a) (eff : a Effect.t) ->
11             let handle_eff : ((a, state -> r * state) continuation -> state -> r
12                 * state) option =
13                 match eff with
14                 | Get -> Some (fun k state -> continue k state state)
15                 | Set x -> Some (fun k _state -> continue k () x)
16                 | Flat (label, check, arg) ->

```

```

16      Some (fun k state ->
17          let (ok, updated_state) = wrap_inner (fun () -> check arg)
18              state in
19              if ok
20              then continue k arg updated_state
21              else discontinue k (Blame label) updated_state)
22      | _ -> None
23  in
24      handle_eff
25  }
26  init
27
28  let wrap main =
29      let r, _ = wrap_inner main state_init in
30      r
31
32  let () = wrap main

```

This approach works, but with a small caveat. Main code can access this state, and can potentially modify the contract's state. This issue is easily resolved by making two more handlers, one handler for state, and one handler perform recursive wrapping for contract checking code. The state handler is installed in the main wrapper, placing on the top most contract checking code. When a contract check occurs it opens a new scope, starting with the current state. All dependent contract checking code originates from this top most contract update on the same state. Once the top most contract code returns, the state is updated at the continuation of the main wrapper. Because the state is registered by the main wrapper, and the handler for state is registered only when contract checking code executes, main code has no access to this state.

However, in this implementation, a `Flat` effect might be invoked arbitrarily by the main code. In an actual language implementation, we suggest this effect be private, or a special keyword to prevent developers interact with the effect state.

```

1  let rec with_state : type res. (unit -> res) -> int -> res * int = 
2    fun thunk init ->
3    match_with thunk ()
4    { retc = (fun x s -> (x, s));
5      exnc = (fun e _ -> raise e);
6      effc = fun (type a) (eff : a Effect.t) ->
7        let handler : ((a, int -> res * int) continuation -> int -> res *
8          int) option =
9          match eff with
10         | Get -> Some (fun k s -> continue k s s)
11         | Set x -> Some (fun k _ -> continue k () x)
12         | _ -> None
13       in handler
14    } init
15
16 let rec wrap_inner : type res. (unit -> res) -> res =
17   fun thunk ->
18   match_with thunk ()
19   { retc = (fun x -> x);
20     exnc = (fun e -> raise e);
21     effc = fun (type a) (eff : a Effect.t) ->
22       match eff with
23       | Flat (label, check, arg) ->
24         Some (fun k ->
25           let ok = wrap_inner (fun () -> check arg) in
26           if ok then continue k arg
27           else discontinue k (Blame label))
28       | _ -> None
29   }
30
31 let rec wrap : type res. (unit -> res) -> int -> res * int =
32   fun thunk init ->
33   match_with thunk ()
34   { retc = (fun x s -> (x, s));
35     exnc = (fun e _ -> raise e);
36     effc = fun (type a) (eff : a Effect.t) ->
37       let handler : ((a, int -> res * int) continuation -> int -> res *
38         int) option =

```

```

37      match eff with
38      | Flat (label, check, arg) ->
39          Some (fun k current_state ->
40              let (ok, next_state) =
41                  with_state (fun () -> wrap_inner (fun () -> check arg))
42                      current_state
43              in
44                  if ok then continue k arg next_state
45                  else discontinue k (Blame label) next_state)
46      | _ -> None
47      in handler
48  } init

```

5.2 Contract Interaction

All contract are unified under a single context. This context is accessible through two effects `Get` and `Set`. Let this context be a simple key-value storage, then programmers can devise complicated schemes to track function calls, current function scope, internal contract state, etc, ...

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CHAPTER A

Base Theorems

Theorem A.1. (Base Type Safety)

- If $\cdot \vdash e_1 : \tau$ and $e_1, \emptyset \longrightarrow e_2, \sigma$ for some σ then $\cdot \vdash e_2 : \tau$.
- If $\cdot \vdash e_1 : \tau$ then e_1 is a value or exists e_2, σ such that $e_1, \emptyset \longrightarrow e_2, \sigma$.

By [Theorem A.2](#) and [Theorem A.3](#).

Theorem A.2. (Base Type Preservation)

If $\cdot \vdash e_1 : \tau$ and $e_1, \emptyset \longrightarrow e_2, \sigma$ for some σ then $\cdot \vdash e_2 : \tau$.

Proof of [Theorem A.2](#).

By induction on $e_1, \sigma \longrightarrow e_2, \sigma'$.

- Case $\lambda x.e_3 \ v, \sigma \longrightarrow e_3[x := v], \sigma$.

$e_1 = \lambda x.e_3 \ v$	Given
$e_2 = e_3[x := v]$	Given
$\cdot \vdash v : \tau_1$	Type-App
$\cdot \vdash \lambda x.e_3 : \tau_1 \rightarrow \tau$	Type-App
$\cdot \vdash e_3 : \tau_1 \rightarrow \tau$	Type-Lambda
$\cdot \vdash x : \tau_1$	Type-Lambda
$\cdot \vdash e_3[x := v] : \tau$	Lemma A.5
$\cdot \vdash e_2 : \tau$	

- Case $\mu x.e_3, \sigma \longrightarrow e_3[x := \mu x.e_3], \sigma$

$e_1 = \mu x.e_3$	Given
$e_2 = e_3[x := \mu x.e_3]$	Given
$\cdot \vdash e_3 : \tau$	Type-Fix
$\cdot \vdash x : \tau$	Type-Fix
$\cdot \vdash e_3[x := \mu x.e_3] : \tau$	Lemma A.5
$\cdot \vdash e_2 : \tau$	

- Case if true then e_3 else $e_4, \sigma \longrightarrow e_3$

$$\begin{array}{ll}
e_1 = \text{if true then } e_3 \text{ else } e_4 & \text{Given} \\
e_2 = e_3 & \text{Given} \\
\cdot \vdash e_3 : \tau & \text{Type-If} \\
\cdot \vdash e_2 : \tau &
\end{array}$$

- Case if false then e_3 else e_4 , $\sigma \longrightarrow e_3$ similarly above.
- Case $\text{inj.l}(\langle v_1, v_2 \rangle), \sigma \longrightarrow v_1$

$$\begin{array}{ll}
e_1 = \text{inj.l}(\langle v_1, v_2 \rangle) & \text{Given} \\
e_2 = v_1 & \text{Given} \\
\cdot \vdash \langle v_1, v_2 \rangle : \langle \tau, \tau_1 \rangle & \text{T-Inj-Left} \\
\cdot \vdash v_1 : \tau & \text{T-Tuple} \\
\cdot \vdash e_2 : \tau &
\end{array}$$

- Case $\text{inj.r}(\langle v_1, v_2 \rangle), \sigma \longrightarrow v_2$ similarly above.
- Case $E[\text{new}(v)], \sigma \longrightarrow E[\text{loc}], \sigma[\text{loc} \rightarrow v]$

$$\begin{array}{ll}
e_1 = E[\text{new}(v)] & \text{Given} \\
e_2 = E[\text{loc}] & \text{Given} \\
\Gamma \vdash E[\text{new}(v)] : \tau & \text{Given} \\
\Gamma \vdash \text{new}(v) : \tau_1 & \text{Decomposition} \\
\Gamma \vdash \text{new}(v) : \text{ref}(\tau_2) & \text{T-New} \\
\Gamma \vdash \text{loc} : \text{ref}(\tau_2) & \text{T-Loc} \\
\Gamma \vdash E[\text{loc}] : \tau & \text{Lemma A.4} \\
\Gamma \vdash e_2 : \tau &
\end{array}$$

- Case $E[\text{get}(\text{loc})], \sigma \longrightarrow E[v], \sigma$ for $v = \sigma(\text{loc})$

$$\begin{array}{ll}
e_1 = E[\text{get}(\text{loc})] & \text{Given} \\
e_2 = E[v] & \text{Given} \\
\Gamma \vdash E[\text{get}(\text{loc})] : \tau & \text{Given} \\
\Gamma \vdash \text{get}(\text{loc}) : \tau_1 & \text{Decomposition} \\
\Gamma \vdash v : \tau_1 & ?? \\
\Gamma \vdash E[v] : \tau & \text{Lemma A.4} \\
\Gamma \vdash e_2 : \tau &
\end{array}$$

- Case $E[\text{set}(v_1, v_2)], \sigma \longrightarrow E[\text{loc}], \sigma[\text{loc} \rightarrow v]$

$e_1 = E[\text{set}(v_1, v_2)]$	Given
$e_2 = E[\text{loc}]$	Given
$\Gamma \vdash E[\text{set}(v_1, v_2)] : \tau$	Given
$\Gamma \vdash \text{set}(v_1, v_2) : \tau_1$	Decomposition
$\Gamma \vdash \text{set}(v_1, v_2) : \text{ref}(\tau_2)$	T-New
$\Gamma \vdash \text{loc} : \text{ref}(\tau_2)$	T-Loc
$\Gamma \vdash E[\text{loc}] : \tau$	Lemma A.4
$\Gamma \vdash e_2 : \tau$	

□

Theorem A.3. (Base Progress)

If $\cdot \vdash e_1 : \tau$ then e_1 is a value or exists e_2, σ such that $e_1, \emptyset \longrightarrow e_2, \sigma$.

Proof of [Theorem A.3](#).

□

Lemma A.4. (Context Replacement)

If $\Gamma \vdash E[e_1] : \tau$ and $\Gamma \vdash e_1 : \tau_1$ then for any e_2 such that, $\Gamma \vdash e_2 : \tau_1$, obtain $\Gamma \vdash E[e_2] : \tau$.

Proof of [Lemma A.4](#). By induction on E .

- Case $E = E_1 e_3$

$E[e_1] = (E_1 e_3)[e_1]$	Given
$E[e_1] = E_1[e_1] e_3$	
$\Gamma \vdash E_1[e_1] e_3 : \tau$	Given
$\Gamma \vdash e_3 : \tau_2$	Type-App
$\Gamma \vdash E_1[e_1] : \tau_2 \rightarrow \tau$	Type-App
$\Gamma \vdash E_1[e_2] : \tau_2 \rightarrow \tau$	IH
$\Gamma \vdash E_1[e_2] e_3 : \tau$	Type-App
$\Gamma \vdash (E_1 e_3)[e_2] : \tau$	
$\Gamma \vdash E[e_2] : \tau$	

- Other cases follow similarly.

□

Lemma A.5. (Substitution)

If $\Gamma \vdash e_1 : \tau_1$ and $\Gamma, x : \tau_1 \vdash e_2 : \tau_2$ then $\Gamma \vdash e_2[x := e_1] : \tau_2$.

Proof of [Lemma A.5](#).

□

CHAPTER B

Contract Erasure

CHAPTER C

Effects Safety

CHAPTER D

Compiler Equivalence for Effect (Deep handlers)

D.1 Overview

Proof for compiler from **Contracts** to Untyped Deep **Effects**, the handler semantic is deep, where the continuation automatically reinstall the handler.

$$\begin{aligned}
 \mathbf{h}_{\mathcal{C}} : \mathbf{lab} &= \{ \mathcal{C} \rightarrow \forall \alpha. \langle \mathbf{blab}, \langle \alpha \rightarrow \text{bool}, \alpha \rangle \rangle \rightarrow \alpha \} \\
 \text{wrap} &= \mu w : \forall \alpha. ((\ () \rightarrow \langle \mathbf{h}_{\mathcal{C}} \rangle \alpha) \rightarrow \alpha). \Lambda \alpha. \lambda f : (\ () \rightarrow \langle \mathbf{h}_{\mathcal{C}} \rangle \alpha). \\
 &\quad \mathbf{handler} \{ \mathcal{C} \rightarrow \lambda \langle l, \langle e, v \rangle \rangle. \lambda k. \text{ if } w[\text{bool}] (\lambda_. e \ v) \text{ then } k \ v \text{ else } \mathbf{error}_l \} f \\
 &\quad \text{where } \lambda \langle l, \langle e, v \rangle \rangle \text{ deconstructs the argument as tuple pattern}
 \end{aligned}$$

The proof get into the way a lot because it switch back and forth between states prepended with handle and states without handle.

Compiler Correctness by proving (1) **Contracts** $\sim$ Untyped **Effects**, (2)
 Untyped **Effects** $\sim$ **Effects**, then obtain **Contracts** $\sim$ **Effects**. (2) is trivial.

D.2 Relation

Let's first define these relations:

- $R_e \subseteq \mathbf{Contracts} \ e \times \text{Untyped } \mathbf{Effects} \ e$

If $(e, e') \in R_e$, we write $e \sim_{R_e} e'$ or $e \sim e'$. We define $\tilde{e}$ as shorthand for any expression $\tilde{e}$ satisfying $e \sim \tilde{e}$.

- $R_E \subseteq \mathbf{Contracts} \ E \times \text{Untyped } \mathbf{Effects} \ E$

If $(E, E') \in R_E$, we write $E \sim_{R_E} E'$ or $E \sim E'$. We define $\tilde{E}$ as shorthand for any context $\tilde{E}$ satisfying $E \sim \tilde{E}$.

We define a separated context to deal with how $\text{check}_j^k(e, v)$ are related from source to target.

$$\sim \subseteq \text{Contracts } E^+ \times \text{Untyped Effects } E^+$$

$$E^+ = \square \mid E^+ [E^* [\text{check}_j^k(\square, v)]]$$

Define $\tilde{E}^+$ such that $E^+ \sim \tilde{E}^+$.

$$\square \sim \square$$

$$E^+ [E^* [\text{check}_j^k(\square, v)]] \sim \tilde{E}^+ [\text{if } \square \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}^*[x]) \tilde{v} \text{ else blame}]$$

$$\sim \subseteq \text{Contracts } E^* \times \text{Untyped Effects } E^*$$

E^* is a context without $\text{check}_j^k(e, v)$ in **Contracts**.

Define $\tilde{E}^*$ such that $E^* \sim \tilde{E}^*$.

$$\square \sim \square$$

$$E^* e \sim \tilde{E}^* \tilde{e}$$

$$v E^* \sim \tilde{v} \tilde{E}^*$$

$$\text{mon}_j^{k,l}(\text{flat}(e), E^*) \sim \text{perform } \mathcal{C} \langle \tilde{e}, \tilde{E}^* \rangle$$

$$\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, E^*) \sim \tilde{e}_1 \tilde{E}^*$$

e_1 as in the $\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, e)$ row below

$$\text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, E^*) \sim \tilde{e}_1 \tilde{E}^*$$

e_1 as in the $\text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, e)$ row

below

$$\sim \subseteq \text{Contracts } e \times \text{Untyped Effects } e$$

E^* is a context without $\text{check}_j^k(e, v)$ in **Contracts**. So it does not have the check cases below.

$$v \sim \tilde{v}$$

$$e_1 e_2 \sim \tilde{e}_1 \tilde{e}_2$$

$$\text{mon}_j^{k,l}(\text{flat}(e_1), e_2) \sim \text{perform } \mathcal{C} \langle \tilde{e}_1, \tilde{e}_2 \rangle$$

$$\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, e) \sim \tilde{e}_1 \tilde{e}$$

$$e_1 = \lambda f. \lambda x. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))$$

$$\begin{aligned}
\text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, e) &\sim \tilde{e}_1 \tilde{e} \\
e_1 &= \lambda f. \lambda x. \text{mon}_j^{k,l}(\{\text{mon}_j^{l,j}(\kappa_1, x)/y\} \kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) \\
\text{blame}_j^k &\sim \text{error}_j
\end{aligned}$$

$$\sim \subseteq \text{Contracts } v \times \text{Untyped Effects } v$$

$$\begin{aligned}
x &\sim x \\
b &\sim b \\
N &\sim N \\
\lambda x. e &\sim \lambda x. \tilde{e} \\
\text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) &\sim \tilde{e} \\
e &= \lambda x. \text{mon}_j^{k,l}(\kappa_2, \tilde{v} \text{mon}_j^{l,k}(\kappa_1, x)) \\
\text{guard}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, v) &\sim \tilde{e} \\
e &= \lambda x. \text{mon}_j^{k,l}(\{\text{mon}_j^{l,j}(\kappa_1, x)/\widetilde{y}\} \kappa_2, v \text{mon}_j^{l,k}(\kappa_1, x))
\end{aligned}$$

Lemma 4.1. (Relation is correct) If $e \sim e'$ and $E \sim E'$ then $E[e] \sim E'[e']$.

Proof. By induction on the structure of E .

Base case: $E = \square$. Then $E[e] = e \sim e' = E'[e']$ by assumption.

Inductive cases:

- Case $E = E_1 e_0$ and $E' = E'_1 e'_0$ where $E_1 \sim E'_1$ and $e_0 \sim e'_0$.

By I.H., $E_1[e] \sim E'_1[e']$. By relation definition, $E_1[e] e_0 \sim E'_1[e'] e'_0$.

- Case $E = v E_1$ and $E' = v' E'_1$ where $v \sim v'$ and $E_1 \sim E'_1$.

By I.H., $E_1[e] \sim E'_1[e']$. By relation definition, $v E_1[e] \sim v' E'_1[e']$.

- Case $E = \text{mon}_j^{k,l}(\text{flat}(e_0), E_1)$ and $E' = \text{perform } \mathcal{C} \langle e'_0, E'_1 \rangle$ where $e_0 \sim e'_0$ and $E_1 \sim E'_1$.

By I.H., $E_1[e] \sim E'_1[e']$. By relation definition, $\text{mon}_j^{k,l}(\text{flat}(e_0), E_1[e]) \sim \text{perform } \mathcal{C} \langle e'_0, E'_1[e'] \rangle$.

- Case $E = \text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, E_1)$ and $E' = e'_c E'_1$ where $e_c \sim e'_c$ and $E_1 \sim E'_1$, with $e_c = \lambda f. \lambda x. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))$.

By I.H., $E_1[e] \sim E'_1[e']$. By the $\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, e)$ row of the relation (with $e := E_1[e]$ and $\tilde{e}_1 := e'_c$), $\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, E_1[e]) \sim e'_c E'_1[e']$.

- Case $E = \text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y.\kappa_2, E_1)$ and $E' = e'_c E'_1$ where $e_c \sim e'_c$ and $E_1 \sim E'_1$, with $e_c = \lambda f.\lambda x.\text{mon}_j^{k,l}(\{\text{mon}_j^{l,j}(\kappa_1, x)/y\}\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))$.

By I.H., $E_1[e] \sim E'_1[e']$. By the $\text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y.\kappa_2, e)$ row of the relation (with $e := E_1[e]$ and $\tilde{e}_1 := e'_c$), $\text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y.\kappa_2, E_1[e]) \sim e'_c E'_1[e']$.

Note: We don't need explicit check cases here since E^* contexts don't contain checks by definition.

All cases follow from the induction hypothesis and the compositional nature of the relation. $\square$

Lemma 4.2. (Relation is closed under substitution) If $e \sim e'$ and $v \sim v'$ then $e[x := v] \sim e'[x := v']$.

Proof. By induction on the derivation of $e \sim e'$.

- Case $x \sim x$. Then $x[x := v] = v \sim v' = x[x := v']$.
- Case $z \sim z$ with $z \neq x$. Both sides are unchanged.
- Case $\lambda z.e_1 \sim \lambda z.e'_1$ with $e_1 \sim e'_1$. Choose $z \neq x$ and $z \notin \text{fv}(v, v')$ by renaming. By I.H., $e_1[x := v] \sim e'_1[x := v']$, then the λ row.
- All other rows (**apply**, **mon** with flat, arrow and dependent contracts, **guard**, **b**, **N**, **blame**) are compositional. The substitution goes into every related sub-expression, including the predicates inside contracts and the wrappers e_1 , and each follows by I.H.

$\square$

Lemma 4.3. (Breaking down Context E) For all E , $E = E^+[E^*]$

Proof. By induction on E .

- Case $E = \square$

Choose $E^+ = E^* = \square$

- Case $E = E_1 e$

By IH, $E_1 = E_1^+[E_1^*]$

$E = E_1^+[E_1^*] e$

- Case $E_1^+ = \square$

Then $E = E_1^* e$

Choose $E^+ = \square, E^* = E_1^* e$

- Case $E_1^+ = E_2^+[E_2^*[\text{check}_j^k(\square, v)]]$

Then $E = E_2^+ [E_2^* [\text{check}_j^k(E_1^*, v)]]$ $e = (E_2^+ e) [E_2^* [\text{check}_j^k(E_1^*, v)]]$

Choose $E^+ = (E_2^+ e) [E_2^* [\text{check}_j^k(\square, v)]]$, $E^* = E_1^*$

Similarly to other basic cases.

- Case $E = \text{check}_j^k(E_1, v)$

By IH, $E_1 = E_1^+ [E_1^*]$

$E = \text{check}_j^k(E_1^+ [E_1^*], v) = \text{check}_j^k(E_1^+, v) [E_1^*]$

- Case $E_1^+ = \square$

Then $E = \text{check}_j^k(\square, v) [E_1^*]$

Let $E^+ = \text{check}_j^k(\square, v)$, $E^* = E_1^*$

- Case $E_1^+ = E_2^+ [E_2^* [\text{check}_{j_1}^{k_1}(\square, v_1)]]$.

Then $E = \text{check}_j^k(E_2^+ [E_2^* [\text{check}_{j_1}^{k_1}(\square, v_1)]] , v) [E_1^*]$

Let $E^+ = \text{check}_j^k(E_2^+ [E_2^* [\text{check}_{j_1}^{k_1}(\square, v_1)]] , v)$, $E^* = E_1^*$

(This last step is because $E[E_0^+] = E_1^+$ Lemma 4.4)

□

Lemma 4.4. (Context composition preserves check structure)

If $E^+ = E_0^+ [E_0^* [\text{check}_j^k(\square, v)]]$ and E is any evaluation context, then $E[E^+] = E_1^+$ for some E_1^+ of the form $E_1^+ = E_2^+ [E_2^* [\text{check}_{j'}^{k'}(\square, v')]]$.

Proof. By induction on the structure of E .

Base case: $E = \square$. Then $E[E^+] = E^+ = E_0^+ [E_0^* [\text{check}_j^k(\square, v)]]$, which is already in the required form.

Inductive cases:

- Case $E = E_1 e$. By I.H., $E_1[E^+] = E_1^+$ where $E_1^+ = E_2^+ [E_2^* [\text{check}_{j'}^{k'}(\square, v')]]$.

Then $E[E^+] = E_1[E^+] e = E_2^+ [E_2^* [\text{check}_{j'}^{k'}(\square, v')]] e$

$= (E_2^+ e) [E_2^* [\text{check}_{j'}^{k'}(\square, v')]]$

Taking $E_3^+ = E_2^+ e$, we get the required form.

- Case $E = v E_1$. Similar to the previous case.
- Case $E = \text{check}_{j_0}^{k_0}(E_1, v_0)$. By I.H., $E_1[E^+] = E_1^+$.

Then $E[E^+] = \text{check}_{j_0}^{k_0}(E_1^+, v_0)$

If $E_1^+ = \square$, then $E[E^+] = \text{check}_{j_0}^{k_0}(\square, v_0)$ which has the required form.

If $E_1^+ = E_2^+ [E_2^* [\text{check}_{j'}^{k'}(\square, v')]]$, then:
 $E[E^+] = \text{check}_{j_0}^{k_0}(E_2^+ [E_2^* [\text{check}_{j'}^{k'}(\square, v')]] , v_0)$
 $= \text{check}_{j_0}^{k_0}(E_2^+, v_0) [E_2^* [\text{check}_{j'}^{k'}(\square, v')]]$

Taking $E_3^+ = \text{check}_{j_0}^{k_0}(E_2^+, v_0)$, we get the required form.

- Other cases (mon contexts) follow similarly.

In all cases, composing any context with a check-containing context yields another check-containing context. $\square$

Lemma 4.5. (Compiler is subset of relation)

If $\Gamma \vdash e_1 : \tau \succrightarrow e_2$ then $e_1 \sim e_2^*$.

In particular, for closed programs, if $\cdot \vdash e_1 : \tau \succrightarrow e_2$ then $e_1 \sim e_2^*$.

In this lemma, we use e^* as shorthand to denote e untyped.

Proof.

By induction on structure of $\Gamma \vdash e_1 : \tau \succrightarrow e_2$.

- Case

$$\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau \succrightarrow x}$$

$e_2^* = x$. Then $x \sim x$ by relation.

- Case

$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2 \succrightarrow e'}{\Gamma \vdash \lambda x : \tau_1. e : \tau_1 \rightarrow \tau_2 \succrightarrow \lambda x : \tau_1. e'}$$

$e_2^* = \lambda x. e'^*$

By I.H. on the premise with context $\Gamma, x : \tau_1$, $e \sim e'^*$.

Then $\lambda x. e \sim \lambda x. e'^*$ by relation.

- Case

$$\frac{\Gamma \vdash e_3 : \tau \rightarrow \text{bool} \succrightarrow e_5 \quad \Gamma \vdash e_4 : \tau \succrightarrow e_6}{\Gamma \vdash \text{mon}_j^{k,l}(\text{flat}(e_3), e_4) : \tau \succrightarrow \text{perform } \mathcal{C} \tau \langle k, \langle e_5, e_6 \rangle \rangle}$$

$e_2^* = \text{perform } \mathcal{C} \langle k, \langle e_5^*, e_6^* \rangle \rangle$

By I.H. $e_3 \sim e_5^*$ and $e_4 \sim e_6^*$.

Then $e_1 \sim e_2^*$ by relation.

$e'_1 \Rightarrow^* b$ by Lemma 4.8.

$e^* \Rightarrow^* b$ by combining steps. $\square$

Lemma 4.7. (Wrapped compiled program steps to an equivalent state with source program)

If $\cdot \vdash e : \tau \mapsto e'$ and $e_1^* = \text{untyped}(\text{wrap}[\tau] (\lambda_.e'))$ then it exists e_2^* such that $e_1^* \Rightarrow^* e_2^*$ and $\text{EQ}(e, e_2^*)$.

In this lemma, we use e^* as shorthand to denote e untyped.

Proof.

By Lemma 4.5, $e \sim e'^*$.

$$e_1^* \Rightarrow^* \text{handler } h_{\mathcal{C}} \lambda_.e'^* \Rightarrow \text{handle } h_{\mathcal{C}} e'^* = e_2^*$$

By definition, $\text{EQ}(e, e_2^*)$. $\square$

Lemma 4.8. If $e \rightarrow^* r$ and $\text{EQ}(e, e')$ then $e' \Rightarrow^* \tilde{r}$.

Proof. By breaking down the $\rightarrow^*$, and induction on it.

- Case $v \rightarrow^* v$

$e' = \tilde{r}$ because $\text{EQ}(e, e')$.

$e' \Rightarrow^* \tilde{r}$. Reflexivity.

- Case $\text{blame}_j^k \rightarrow^* \text{error}_j$

$e' = \tilde{r}$ because $\text{EQ}(e, e')$.

$e' \Rightarrow^* \tilde{r}$. Reflexivity.

- Case $e \rightarrow e_1$ and $e_1 \rightarrow^* r$

By Lemma 4.9, we have $e' \Rightarrow^* e'_1$ such that $\text{EQ}(e_1, e'_1)$.

By IH, $e'_1 \Rightarrow^* \tilde{r}$.

By chaining steps, $e' \Rightarrow^* \tilde{r}$. $\square$

We also need to make a claim that in no where in e'_1 do we have **perform eff** that **eff** $\neq \mathcal{C}$.

Lemma 4.9. If $e_1 \rightarrow e_2$ and $\text{EQ}(e_1, e'_1)$ then there exists e'_2 such that $e'_1 \Rightarrow^* e'_2$ and $\text{EQ}(e_2, e'_2)$.

Proof of Lemma 4.9. By case analysis on $e_1 \rightarrow e_2$ with respect to $\text{EQ}(e_1, e'_1)$. By decomposition lemma, we know that the program can be rewritten into:

$e_1 = E[e_3]$, $e_2 = E[e_4]$ and that $e_3 \rightarrow e_4$ is a redex reduction.

- Case $e_3 = \text{mon}_j^{k,l}(\text{flat}(e), v) \rightarrow \text{check}_j^k(e\ v, v) = e_4$
 - Case $e_1 = v$ is invalid.

Context breakdown lemma, $E = E^+[E^*]$.

$$e_1 = E^+[E^*[e_3]]$$

- Case $E^+ = \square$

$$e_1 = E^*[e_3] \sim \tilde{E}^*[\text{perform } \mathcal{C} \langle \tilde{e}, \tilde{v} \rangle]$$

$$\text{By EQ}(e_1, e'_1) \text{ (case 2), } e'_1 = \text{handle } h_{\mathcal{C}} \tilde{E}^*[\text{perform } \mathcal{C} \langle \tilde{e}, \tilde{v} \rangle]$$

$$e'_1 \Rightarrow^* \text{if } \text{handle } h_{\mathcal{C}} \tilde{e} \tilde{v} \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}^*[x]) \tilde{v} \text{ else } \text{error}_j = e'_2$$

$$\text{By case 2, EQ}(e\ v, \text{handle } h_{\mathcal{C}} \tilde{e} \tilde{v})$$

Immediately $\text{EQ}(e_2, e'_2)$ by case 3.

- Case $E^+ = E_1^+[E_1^*[\text{check}_{j_1}^{k_1}(\square, v_1)]]$

$$e_1 = E_1^+[E_1^*[\text{check}_{j_1}^{k_1}(E^*[e_3], v_1)]]$$

$$e'_1 = \tilde{E}_1^+[\text{if } \text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{e}_3] \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}_1^*[x]) \tilde{v}_1 \text{ else } \text{error}_{j_1}]$$

$$= \tilde{E}_1^+[\text{if } \text{handle } h_{\mathcal{C}} \tilde{E}^*[\text{perform } \mathcal{C} \langle \tilde{e}, \tilde{v} \rangle] \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}_1^*[x]) \tilde{v}_1 \text{ else } \text{error}_{j_1}]$$

$$\Rightarrow^* \tilde{E}^+[\text{if } \text{handle } h_{\mathcal{C}} \tilde{e} \tilde{v} \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}^*[x]) \tilde{v} \text{ else } \text{error}_j] = e'_2$$

$$\text{where } \tilde{E}^+ = \tilde{E}_1^+[\text{if } \square \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}_1^*[x]) \tilde{v}_1 \text{ else } \text{error}_{j_1}]$$

$$e_2 = E_1^+[E_1^*[\text{check}_{j_1}^{k_1}(E^*[\text{check}_j^k(e\ v, v)], v_1)]]$$

$$\text{EQ}(E^*[\text{check}_j^k(e\ v, v)], \text{if } \text{handle } h_{\mathcal{C}} \tilde{e} \tilde{v} \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}^*[x]) \tilde{v} \text{ else } \text{error}_j) \\ \text{(case 3)}$$

By case 3, $\text{EQ}(e_2, e'_2)$.

(Case 2 is invalid, because of the shape of e_2)

- Case $e_3 = \text{check}_j^k(\text{true}, v) \rightarrow v = e_4$
 - Case $e_1 = v$ is invalid.
 - Case $e_1 = E^+[E^*[\text{check}_j^k(\text{true}, v)]]$.

By $\text{EQ}(e_1, e'_1)$:

$$e'_1 = \tilde{E}^+[\text{if } e'_3 \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}^*[x]) \tilde{v} \text{ else error}_{j_1}]$$

EQ(true, e'_3), because true is a value, we obtain $e'_3 = \text{true}$.

$$e'_1 \Rightarrow^* \tilde{E}^+[\text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{v}]]$$

$$e_2 = E^+[E^*[v]]$$

– Case $E^+ = \square$

Straight up, EQ($E^*[v]$, $\text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{v}]$)

– Case $E^+ = E_1^+[E_1^*[\text{check}_{j_1}^{k_1}(\square, v_1)]]$

$$\tilde{E}^+ = \tilde{E}_1^+[\text{if } \square \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}_1^*[x]) \tilde{v}_1 \text{ else error}_{j_1}]$$

$$e_2 = E_1^+[E_1^*[\text{check}_{j_1}^{k_1}(E^*[v], v_1)]]$$

$$e'_2 = \tilde{E}_1^+[\text{if } \text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{v}] \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}_1^*[x]) \tilde{v}_1 \text{ else error}_{j_1}]$$

Trivially, EQ($E^*[v]$, $\text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{v}]$) (case 2).

Obtain EQ(e_2, e'_2) (case 3).

• Case $e_3 = \text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) \rightarrow \text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) = e_4$

▸ Case $e_1 = v$ is invalid.

Context breakdown lemma, $E = E^+[E^*]$.

$$e_1 = E^+[E^*[e_3]]$$

▸ Case $E^+ = \square$

$$\text{Let } e_5 = \lambda f. \lambda x. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) = \lambda f. \lambda x. e_{5a}$$

$$e_1 = E^*[e_3] \sim \tilde{E}^*[(\lambda f. \lambda x. \tilde{e}_{5a}) \tilde{v}]$$

$$\text{By EQ}(e_1, e'_1) \text{ (case 2), } e'_1 = \text{handle } h_{\mathcal{C}} \tilde{E}^*[(\lambda f. \lambda x. \tilde{e}_{5a}) \tilde{v}]$$

$$e'_1 \Rightarrow^* \text{handle } h_{\mathcal{C}} \tilde{E}^*[\lambda x. \tilde{e}_{5a}[f := \tilde{v}]] = e'_2$$

$$e_2 = E^*[\text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v)]$$

$$\text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) \sim \lambda x. \tilde{e}_{5a}[f := \tilde{v}] \text{ by the relation on values.}$$

Obtain EQ(e_2, e'_2) (case 2).

▸ Case $E^+ = E_1^+[E_1^*[\text{check}_{j_1}^{k_1}(\square, v_1)]]$

$$\text{Let } e_5 = \lambda f. \lambda x. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))$$

$$e'_1 = \tilde{E}_1^+[\text{if } \text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{e}_5 \tilde{v}] \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}_1^*[x]) \tilde{v}_1 \text{ else error}_{j_1}]$$

$$e'_1 \Rightarrow^* \tilde{E}_1^+[\text{if } \text{handle } h_{\mathcal{C}} \tilde{E}^*[\lambda x. \tilde{e}_{5a}[f := \tilde{v}]] \text{ then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}_1^*[x]) \tilde{v}_1 \text{ else error}_{j_1}] = e'_2$$

where $e_{5a} = \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))$

$$e_2 = E_1^+ [E_1^* [\text{check}_{j_1}^{k_1}(E^* [\text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v)], v_1)]]$$

Since $\text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) \sim \lambda x. \tilde{e}_{5a}[f := \tilde{v}]$, we have $\text{EQ}(e_2, e'_2)$ (case 3).

- Case $e_3 = \text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v_1) v_2 \rightarrow \text{mon}_j^{k,l}(\kappa_2, v_1 \text{mon}_j^{l,k}(\kappa_1, v_2)) = e_4$
 - Case $e_1 = v$ is invalid.

Context breakdown lemma, $E = E^+[E^*]$.

$$e_1 = E^+[E^*[e_3]]$$

- Case $E^+ = \square$

$$\tilde{e}_3 = \tilde{e}_5 \tilde{v}_2 \text{ where } e_5 = \lambda x. \text{mon}_j^{k,l}(\kappa_2, \tilde{v}_1 \text{mon}_j^{l,k}(\kappa_1, x))$$

By $\text{EQ}(e_1, e'_1)$ (case 2), $e'_1 = \text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{e}_3]$

$$e'_1 \Rightarrow^* \text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{e}_4] = e'_2 \text{ by } \beta.$$

$$e_2 = E^*[e_4]$$

Obtain $\text{EQ}(e_2, e'_2)$ (case 2).

- Case $E^+ = E_1^+ [E_1^* [\text{check}_{j_1}^{k_1}(\square, v_1)]]$

Same as the case $E^+ = \square$, inside $\tilde{E}^+$, by case 3.

- Case $e_3 = \text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, v) \rightarrow \text{guard}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, v) = e_4$

Same as the case $\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v)$, with

$$e_5 = \lambda f. \lambda x. \text{mon}_j^{k,l}(\{\text{mon}_j^{l,j}(\kappa_1, x)/y\} \kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) = \lambda f. \lambda x. e_{5a}$$

$$\lambda x. \tilde{e}_{5a}[f := \tilde{v}] \sim \text{guard}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, v) \text{ by the relation on values.}$$

- Case $e_3 = \text{guard}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, v_1) v_2 \rightarrow \text{mon}_j^{k,l}(\kappa_3, v_1 \text{mon}_j^{l,k}(\kappa_1, v_2)) = e_4$ where $\kappa_3 = \{\text{mon}_j^{l,j}(\kappa_1, v_2)/y\} \kappa_2$

- Case $e_1 = v$ is invalid.

Context breakdown lemma, $E = E^+[E^*]$.

Let $e_6 = \text{mon}_j^{k,l}(\{\text{mon}_j^{l,j}(\kappa_1, x)/y\} \kappa_2, v_1 \text{mon}_j^{l,k}(\kappa_1, x))$. Since $x \notin \text{fv}(v_1, \kappa_1, \kappa_2)$, $e_4 = e_6[x := v_2]$.

- Case $E^+ = \square$

$\tilde{e}_3 = (\lambda x. \tilde{e}_6) \tilde{v}_2$ by the guard row of the relation on values.

By $\text{EQ}(e_1, e'_1)$ (case 2), $e'_1 = \text{handle } h_{\mathcal{C}} \tilde{E}^*[(\lambda x. \tilde{e}_6) \tilde{v}_2]$

$$e'_1 \Rightarrow^* \text{handle } h_{\mathcal{C}} \tilde{E}^*[\tilde{e}_6[x := \tilde{v}_2]] = e'_2 \text{ by } \beta.$$

$$e_2 = E^*[e_6[x := v_2]]$$

$$e_6[x := v_2] \sim \tilde{e}_6[x := \tilde{v}_2] \text{ by Lemma 4.2.}$$

Obtain $\text{EQ}(e_2, e'_2)$ (case 2).

- Case $E^+ = E_1^+[E_1^*[\text{check}_{j_1}^{k_1}(\square, v_1)]]$

Same as the case $E^+ = \square$, inside $\tilde{E}^+$, by case 3.

The indy monitors $\text{mon}_j^{l,j}(\kappa_1, v_2)$ inside κ_3 are ordinary **mon** redexes. Each one is related to the compiled monitor at the same position in $\tilde{e}_6[x := \tilde{v}_2]$, so when the source evaluates one, it is one of the **mon** cases above.

- Case $e_3 = \text{check}_j^k(\text{false}, v) \rightarrow \text{blame}_j^k = e_4$
- Case $e_1 = E^+[E^*[\text{check}_j^k(\text{false}, v)]]$

By $\text{EQ}(e_1, e'_1)$:

$$e'_1 = \tilde{E}^+[\text{if false then } (\lambda x. \text{handle } h_{\mathcal{C}} \tilde{E}^*[x]) \tilde{v} \text{ else } \text{error}_j]$$

$$e'_1 \Rightarrow^* \tilde{E}^+[\text{error}_j] = e'_2$$

$$e_2 = E^+[E^*[\text{blame}_j^k]]$$

By Step-E-Error rule: $\tilde{E}^+[\text{error}_j] \Rightarrow \text{error}_j$

Since $\text{blame}_j^k \sim \text{error}_j$, we have $\text{EQ}(e_2, e'_2)$.

□

CHAPTER E

Compiler Equivalence for Effect (Shallow handlers)

E.1 Overview

Proof for compiler from **Contracts** to Untyped Shallow **Effects**, the handler semantic is shallow, where the continuation does not reinstall the handler.

The only rule that changes from the deep semantics is Step-E-Perform, the continuation k no longer wraps E with **handle** h .

$$\frac{\text{op} \notin \mathbf{bop}(E) \wedge (\text{op} \rightarrow f) \in h \quad k = \lambda x. E[x]}{\mathbf{handle} \ h \ E[\mathbf{perform} \ \text{op} \ v] \rightarrow f \ v \ k} \text{Step-E-Perform-Shallow}$$

$$\mathbf{h}_{\mathcal{C}} : \mathbf{lab} = \{ \mathcal{C} \rightarrow \forall \alpha. \langle \mathbf{blab}, \langle \alpha \rightarrow \text{bool}, \alpha \rangle \rangle \rightarrow \alpha \}$$

$$\mathbf{H} = \mu h. \mathbf{handler} \{ \mathcal{C} \rightarrow \lambda \langle l, \langle e, v \rangle \rangle. \lambda k. h \ (\lambda_. \text{ if } e \ v \text{ then } k \ v \text{ else } \mathbf{error}_l) \}$$

$$\mathbf{wrap} = \Lambda \alpha. \lambda f : \text{unit} \rightarrow \alpha. \mathbf{H} \ f$$

where $\lambda \langle l, \langle e, v \rangle \rangle$ deconstructs the argument as tuple pattern

The handler clause re-installs $\mathbf{H}$ around the rest of the program by itself, instead of the continuation doing it. So the whole program always runs under exactly one **handle** $\mathbf{h}_{\mathcal{C}}$.

at the top, and the predicate of a check runs under the same handler as the program. We write **handle** $\mathbf{h}_{\mathcal{C}} \ e$ for the program running under the handler of $\mathbf{H}$.

Compared to the deep semantics, there are no states without handle, and there is no nested handle for predicates. This makes the relation much simpler.

Compiler Correctness by proving (1) **Contracts** $\sim$ Untyped **Effects**, (2)

Untyped **Effects** $\sim$ **Effects**, then obtain **Contracts** $\sim$ **Effects**. (2) is trivial.

In this proof, we use e^* as shorthand to denote e untyped.

Assumptions:

- Contract predicates are values ($\text{flat}(e)$ with $e = \lambda x.e_0$) and pure (they do not use the store).
- Tuple contracts and mutable cells are not covered.

E.2 Relation

Let's first define these relations:

- $R_e \subseteq \text{Contracts } e \times \text{Untyped Effects } e$

If $(e, e') \in R_e$, we write $e \sim_{R_e} e'$ or $e \sim e'$. We define $\tilde{e}$ as shorthand for any expression $\tilde{e}$ satisfying $e \sim \tilde{e}$.

- $R_E \subseteq \text{Contracts } E \times \text{Untyped Effects } E$

If $(E, E') \in R_E$, we write $E \sim_{R_E} E'$ or $E \sim E'$. We define $\tilde{E}$ as shorthand for any context $\tilde{E}$ satisfying $E \sim \tilde{E}$.

In the shallow semantics, the innermost check captures everything outside of it as its continuation, so a check is related together with its whole context:

$$E[\text{check}_j^k(e, v)] \sim \text{if } \tilde{e} \text{ then } (\lambda x. \widetilde{E[x]}) \tilde{v} \text{ else } \text{error}_k$$

where e has no check in evaluation position. E may contain other checks, $\widetilde{E[x]}$ is again related by this rule, and it has one less check.

$$\text{EQ}(e, e') = \begin{cases} \text{true if } e' = \text{handle } h_{\mathcal{C}} \tilde{e} \\ \text{true if } e = E[\text{blame}_j^k] \text{ and } e' = \text{handle } h_{\mathcal{C}} \text{error}_k \\ \text{false otherwise} \end{cases}$$

$\sim \subseteq \text{Contracts } E \times \text{Untyped Effects } E$

Define $\tilde{E}$ such that $E \sim \tilde{E}$. If E contains a check, only the last row applies, on the innermost check.

$$\begin{aligned} \square & \sim \square \\ E e & \sim \tilde{E} \tilde{e} \\ v E & \sim \tilde{v} \tilde{E} \\ \text{mon}_j^{k,l}(\text{flat}(e), E) & \sim \text{perform } \mathcal{C} \langle k, \langle \tilde{e}, \tilde{E} \rangle \rangle \end{aligned}$$

$$\begin{aligned}
\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, E) &\sim \tilde{e}_1 \tilde{E} \\
&e_1 \text{ as in the } \text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, e) \text{ row below} \\
\text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, E) &\sim \tilde{e}_1 \tilde{E} \\
&e_1 \text{ as in the } \text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, e) \text{ row} \\
&\text{below} \\
E[\text{check}_j^k(E_1, v)] &\sim \text{if } \tilde{E}_1 \text{ then } (\lambda x. \widetilde{E[x]}) \tilde{v} \text{ else } \text{error}_k \\
&\text{where } E_1 \text{ has no check}
\end{aligned}$$

The other contexts $(\langle E, e \rangle, o(v \dots, E, e \dots), \text{if } E \text{ then } e_1 \text{ else } e_2, \dots)$ are related homomorphically.

$\sim \subseteq \text{Contracts } e \times \text{Untyped Effects } e$

$$\begin{aligned}
v &\sim \tilde{v} \\
e_1 e_2 &\sim \tilde{e}_1 \tilde{e}_2 \\
\text{mon}_j^{k,l}(\text{flat}(e_1), e_2) &\sim \text{perform } \mathcal{C} \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle \rangle \\
\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, e) &\sim \tilde{e}_1 \tilde{e} \\
&e_1 = \lambda f. \lambda x. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) \\
\text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, e) &\sim \tilde{e}_1 \tilde{e} \\
&\lambda y. \kappa_2, e) \\
&e_1 = \lambda f. \lambda x. (\lambda y. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))) \text{mon}_j^{l,j}(\kappa_1, x) \\
E[\text{check}_j^k(e, v)] &\sim \text{if } \tilde{e} \text{ then } (\lambda x. \widetilde{E[x]}) \tilde{v} \text{ else } \text{error}_k \\
&\text{where } e \text{ has no check in evaluation position (innermost check)} \\
\{e_y/y\}e &\sim \tilde{e}[y := \tilde{v}_y] \\
&\text{where } e_y = \text{mon}_j^{l,j}(\kappa_1, v), e_y \rightarrow^* v_y \text{ and } v_y \sim \tilde{v}_y \text{ (indy argument)} \\
\text{blame}_j^k &\sim \text{error}_k
\end{aligned}$$

The other expressions $(\langle e_1, e_2 \rangle, o(e \dots), \text{if } e_1 \text{ then } e_2 \text{ else } e_3, \mu x. e, \dots)$ are related homomorphically.

$\sim \subseteq \text{Contracts } v \times \text{Untyped Effects } v$

$$\begin{aligned}
b &\sim b \\
N &\sim N \\
\lambda x.e &\sim \lambda x.\tilde{e} \\
\text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) &\sim \tilde{e} \\
e &= \lambda x.\text{mon}_j^{k,l}(\kappa_2, \tilde{v} \text{mon}_j^{l,k}(\kappa_1, x)) \\
\text{guard}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y.\kappa_2, v) &\sim \tilde{e} \\
e &= \lambda x.(\lambda y.\text{mon}_j^{k,l}(\kappa_2, \tilde{v} \text{mon}_j^{l,k}(\kappa_1, x))) \text{mon}_j^{l,j}(\kappa_1, x)
\end{aligned}$$

Lemma 5.1. (Relation is correct) If $E \sim \tilde{E}$, $e \sim \tilde{e}$ and e has no check in evaluation position, then $E[e] \sim \tilde{E}[\tilde{e}]$.

Proof. By case on E .

- Case E has no check

By induction on E , every row of $E \sim \tilde{E}$ matches a row of $e \sim \tilde{e}$ with the hole filled.

- Case $E = E_0[\text{check}_j^k(E_1, v)]$ where E_1 has no check

$E[e] = E_0[\text{check}_j^k(E_1[e], v)]$ and $E_1[e]$ has no check in evaluation position.

$E[e] \sim$ if $\widetilde{E_1[e]}$ then $(\lambda x.\widetilde{E_0[x]}) \tilde{v}$ else **error**_k by the check row.

$\widetilde{E_1[e]} = \tilde{E}_1[\tilde{e}]$ by the previous case.

$= \tilde{E}[\tilde{e}]$ by the check row of $E \sim \tilde{E}$.

□

Lemma 5.2. (Relation is closed under substitution) If $e \sim \tilde{e}$ and $v \sim \tilde{v}$ then $e[x := v] \sim \tilde{e}[x := \tilde{v}]$.

Proof. By induction on $e \sim \tilde{e}$. Every row is homomorphic on x , and the bound variables f, x, y in the rows are chosen fresh. □

Lemma 5.3. (Compiler is subset of relation)

If $\cdot \vdash e_1 : \tau \multimap e_2$ then $e_1 \sim e_2^*$.

Proof.

By induction on structure of $\cdot \vdash e_1 : \tau \multimap e_2$.

- Case

$$\frac{\cdot \vdash e_3 : \tau \rightarrow \text{bool} \succrightarrow e_5 \quad \cdot \vdash e_4 : \tau \succrightarrow e_6}{\cdot \vdash \text{mon}_j^{k,l}(\text{flat}(e_3), e_4) : \tau \succrightarrow \text{perform } \mathcal{C} \tau \langle k, \langle e_5, e_6 \rangle \rangle}$$

$$e_2^* = \text{perform } \mathcal{C} \langle k, \langle e_5^*, e_6^* \rangle \rangle$$

By I.H. $e_3 \sim e_5^*$ and $e_4 \sim e_6^*$.

Then $e_1 \sim e_2^*$ by relation.

- Case

$$\frac{\cdot \vdash e_3 : \tau_1 \rightarrow \tau_2 \succrightarrow e_4 \quad \cdot \vdash \lambda f. \lambda x. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) : (\tau_1 \rightarrow \tau_2) \rightarrow (\tau_1 \rightarrow \tau_2) \succrightarrow e_5}{\cdot \vdash \text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, e_3) : \tau_1 \rightarrow \tau_2 \succrightarrow e_5 e_4}$$

By I.H., $e_3 \sim e_4^*$

By I.H., $\lambda f. \lambda x. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) \sim e_5^*$

Then $e_1 \sim e_2^*$ by relation.

- Case

$$\frac{\cdot \vdash e_3 : \tau_1 \rightarrow \tau_2 \succrightarrow e_4 \quad \cdot \vdash \lambda f. \lambda x. (\lambda y. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))) \text{mon}_j^{l,j}(\kappa_1, x) : (\tau_1 \rightarrow \tau_2) \rightarrow (\tau_1 \rightarrow \tau_2) \succrightarrow e_5}{\cdot \vdash \text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, e_3) : \tau_1 \rightarrow \tau_2 \succrightarrow e_5 e_4}$$

By I.H., $e_3 \sim e_4^*$

By I.H., $\lambda f. \lambda x. (\lambda y. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))) \text{mon}_j^{l,j}(\kappa_1, x) \sim e_5^*$

Then $e_1 \sim e_2^*$ by relation.

- Other cases are homomorphic, by I.H.

□

E.3 Compiler Correctness

Theorem 5.4. If $\cdot \vdash e : \tau \succrightarrow e'$ and $e^* = \text{untyped}(\text{wrap}[\tau] (\lambda_. e'))$ and $e \rightarrow^* r$ then $e^* \Rightarrow^*$

$\tilde{r}$.

Where r is a value v or blame_j^k .

Proof.

$e^* \Rightarrow^* e'_1$ and $\text{EQ}(e, e'_1)$ by Lemma 5.5.

$e'_1 \Rightarrow^* \tilde{r}$ by Lemma 5.6.

$e^* \Rightarrow^* \tilde{r}$ by combining steps. $\square$

Lemma 5.5. (Wrapped compiled program steps to an equivalent state with source program)

If $\cdot \vdash e : \tau \multimap e'$ and $e_1^* = \text{untyped}(\text{wrap}[\tau](\lambda_.e'))$ then it exists e_2^* such that $e_1^* \Rightarrow^* e_2^*$ and $\text{EQ}(e, e_2^*)$.

Proof.

By Lemma 5.3, $e \sim e'^*$.

$$e_1^* \Rightarrow^* H \lambda_.e'^* \Rightarrow^* \text{handler } h_{\mathcal{C}} \lambda_.e'^* \Rightarrow \text{handle } h_{\mathcal{C}} e'^* = e_2^*$$

By definition, $\text{EQ}(e, e_2^*)$ (case 1). $\square$

Lemma 5.6. If $e \rightarrow^* r$ and $\text{EQ}(e, e')$ then $e' \Rightarrow^* \tilde{r}$.

Proof. By breaking down the $\rightarrow^*$, and induction on it.

- Case $r \rightarrow^* r$ with $r = v$

$e' = \text{handle } h_{\mathcal{C}} \tilde{v}$ because $\text{EQ}(v, e')$ (case 2 is invalid).

$e' \Rightarrow \tilde{v}$ by Step-E-Return.

- Case $r \rightarrow^* r$ with $r = \text{blame}_j^k$

$e' = \text{handle } h_{\mathcal{C}} \text{error}_k$ in both cases of treq .

$e' \Rightarrow \text{error}_k$ by Step-E-Error.

- Case $e \rightarrow e_1$ and $e_1 \rightarrow^* r$

By simulation, we have $e' \Rightarrow^* e'_1$ such that $\text{EQ}(e_1, e'_1)$.

By IH, $e'_1 \Rightarrow^* \tilde{r}$.

By chaining steps, $e' \Rightarrow^* \tilde{r}$. $\square$

We also need to make a claim that in no where in $\tilde{e}$ do we have **handle** or **perform eff** that **eff** $\neq \mathcal{C}$, so $\mathcal{C} \notin \text{bop}(E')$ for every target context E' under the top handle.

Lemma 5.7. (Performing a check) If E' has no **handle**, then

$$\begin{aligned} & \text{handle } h_{\mathcal{C}} E' [\text{perform } \mathcal{C} \langle k, \langle \tilde{e}, \tilde{v} \rangle \rangle] \\ & \Rightarrow^* \text{handle } h_{\mathcal{C}} \text{ if } \tilde{e} \tilde{v} \text{ then } (\lambda x. E'[x]) \tilde{v} \text{ else } \text{error}_k \end{aligned}$$

Proof.

$$\begin{aligned}
& \text{handle } h_{\mathcal{C}} E' [\text{perform } \mathcal{C} \langle k, \langle \tilde{e}, \tilde{v} \rangle \rangle] \\
& \Rightarrow (\lambda \langle l, \langle e, v \rangle \rangle. \lambda k. H (\lambda_. \text{if } e \ v \text{ then } k \ v \text{ else } \text{error}_l)) \langle k, \langle \tilde{e}, \tilde{v} \rangle \rangle (\lambda x. E'[x]) \quad (\text{Step-E-Perform-Shallow}) \\
& \Rightarrow^* H (\lambda_. \text{if } \tilde{e} \ \tilde{v} \text{ then } (\lambda x. E'[x]) \ \tilde{v} \text{ else } \text{error}_k) \\
& \Rightarrow^* \text{handle } h_{\mathcal{C}} \text{ if } \tilde{e} \ \tilde{v} \text{ then } (\lambda x. E'[x]) \ \tilde{v} \text{ else } \text{error}_k \quad (\text{unfold } \mu, \text{Step-E-Handler, } \beta)
\end{aligned}$$

□

Lemma 5.8. If $e_1 \rightarrow e_2$ and $\text{EQ}(e_1, e'_1)$ then there exists e'_2 such that $e'_1 \Rightarrow^* e'_2$ and $\text{EQ}(e_2, e'_2)$.

Proof of Lemma 5.8. By case analysis on $e_1 \rightarrow e_2$ with respect to $\text{EQ}(e_1, e'_1)$.

- Case $e_1 = E[\text{blame}_j^k] \rightarrow \text{blame}_j^k = e_2$ (Step-C-Blame)
 - Case $\text{EQ}(e_1, e'_1)$ is case 2, $e'_1 = \text{handle } h_{\mathcal{C}} \text{error}_k$.
 $e'_2 = e'_1$, and $\text{EQ}(\text{blame}_j^k, e'_2)$ (case 2 with $E = \square$).
 - Case $\text{EQ}(e_1, e'_1)$ is case 1, $e'_1 = \text{handle } h_{\mathcal{C}} \tilde{E}[\text{error}_k]$.
 $e'_1 \Rightarrow \text{handle } h_{\mathcal{C}} \text{error}_k = e'_2$ by Step-E-Error.
 Obtain $\text{EQ}(\text{blame}_j^k, e'_2)$ (case 2 with $E = \square$).

Otherwise, e_1 is not blamed and $\text{EQ}(e_1, e'_1)$ is case 1. By decomposition lemma, $e_1 = E[e_3]$, $e_2 = E[e_4]$ and that $e_3 \rightarrow e_4$ is a redex reduction.

If e_3 is not a check, $e'_1 = \text{handle } h_{\mathcal{C}} \tilde{E}[\tilde{e}_3]$ by Lemma 5.1.

Note that $\tilde{E}$ is an evaluation context of **Effects** without **handle**. Unlike the deep proof, there is no case split on the checks in E , the redex steps under the same top handle.

- Case $e_3 = \text{mon}_j^{k,l}(\text{flat}(e), v) \rightarrow \text{check}_j^k(e \ v, v) = e_4$

$$\tilde{e}_3 = \text{perform } \mathcal{C} \langle k, \langle \tilde{e}, \tilde{v} \rangle \rangle$$

$$e'_1 \Rightarrow^* \text{handle } h_{\mathcal{C}} \text{ if } \tilde{e} \ \tilde{v} \text{ then } (\lambda x. \tilde{E}[x]) \ \tilde{v} \text{ else } \text{error}_k = e'_2 \text{ by Lemma 5.7.}$$

$$e_2 = E[\text{check}_j^k(e \ v, v)], \text{ and } e \ v \text{ has no check in evaluation position.}$$

$$e_2 \sim \text{if } \tilde{e} \ \tilde{v} \text{ then } (\lambda x. \widetilde{E[x]}) \ \tilde{v} \text{ else } \text{error}_k \text{ by the check row.}$$

$$\widetilde{E[x]} = \tilde{E}[x] \text{ by Lemma 5.1.}$$

$$\text{Obtain } \text{EQ}(e_2, e'_2) \text{ (case 1).}$$
- Case $e_3 = \text{check}_j^k(\text{true}, v) \rightarrow v = e_4$

$e_1 = E[\text{check}_j^k(\text{true}, v)]$, the check is the innermost one.

$e'_1 = \text{handle } h_{\mathcal{C}} \text{ if true then } (\lambda x. \widetilde{E[x]}) \tilde{v} \text{ else } \text{error}_k \text{ by the check row.}$

$e'_1 \Rightarrow^* \text{handle } h_{\mathcal{C}} \widetilde{E[x]}[x := \tilde{v}] = e'_2$

$e_2 = E[v] = E[x][x := v] \sim \widetilde{E[x]}[x := \tilde{v}]$ by Lemma 5.2.

Obtain $\text{EQ}(e_2, e'_2)$ (case 1).

- Case $e_3 = \text{check}_j^k(\text{false}, v) \rightarrow \text{blame}_j^k = e_4$

$e_1 = E[\text{check}_j^k(\text{false}, v)]$, the check is the innermost one.

$e'_1 = \text{handle } h_{\mathcal{C}} \text{ if false then } (\lambda x. \widetilde{E[x]}) \tilde{v} \text{ else } \text{error}_k \text{ by the check row.}$

$e'_1 \Rightarrow \text{handle } h_{\mathcal{C}} \text{error}_k = e'_2$

$e_2 = E[\text{blame}_j^k]$

Obtain $\text{EQ}(e_2, e'_2)$ (case 2).

- Case $e_3 = \text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) \rightarrow \text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) = e_4$

Let $e_5 = \lambda f. \lambda x. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x)) = \lambda f. \lambda x. e_{5a}$

$\tilde{e}_3 = \tilde{e}_5 \tilde{v}$

$e'_1 \Rightarrow \text{handle } h_{\mathcal{C}} \tilde{E}[\lambda x. \tilde{e}_{5a}[f := \tilde{v}]] = e'_2$ by β .

$\text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v) \sim \lambda x. \tilde{e}_{5a}[f := \tilde{v}]$ by the relation on values.

Obtain $\text{EQ}(e_2, e'_2)$ (case 1) by Lemma 5.1.

- Case $e_3 = \text{mon}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, v) \rightarrow \text{guard}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, v) = e_4$

Same as the case $\text{mon}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v)$, with

$e_5 = \lambda f. \lambda x. (\lambda y. \text{mon}_j^{k,l}(\kappa_2, f \text{mon}_j^{l,k}(\kappa_1, x))) \text{mon}_j^{l,j}(\kappa_1, x) = \lambda f. \lambda x. e_{5a}$

- Case $e_3 = \text{guard}_j^{k,l}(\kappa_1 \rightarrow \kappa_2, v_1) v_2 \rightarrow \text{mon}_j^{k,l}(\kappa_2, v_1 \text{mon}_j^{l,k}(\kappa_1, v_2)) = e_4$

$\tilde{e}_3 = \tilde{e}_5 \tilde{v}_2$ where $e_5 = \lambda x. \text{mon}_j^{k,l}(\kappa_2, \tilde{v}_1 \text{mon}_j^{l,k}(\kappa_1, x))$

$e'_1 \Rightarrow \text{handle } h_{\mathcal{C}} \tilde{E}[\tilde{e}_4] = e'_2$ by β and Lemma 5.2.

Obtain $\text{EQ}(e_2, e'_2)$ (case 1) by Lemma 5.1.

- Case $e_3 = \text{guard}_j^{k,l}(\kappa_1 \rightarrow_i \lambda y. \kappa_2, v_1) v_2 \rightarrow \text{mon}_j^{k,l}(\kappa_3, v_1 \text{mon}_j^{l,k}(\kappa_1, v_2)) = e_4$ where $\kappa_3 = \{\text{mon}_j^{l,j}(\kappa_1, v_2)/y\} \kappa_2$

$\tilde{e}_3 = \tilde{e}_5 \tilde{v}_2$ where $e_5 = \lambda x. (\lambda y. e_{5a}) \text{mon}_j^{l,j}(\kappa_1, x)$ and $e_{5a} = \text{mon}_j^{k,l}(\kappa_2, \tilde{v}_1 \text{mon}_j^{l,k}(\kappa_1, x))$

$e'_1 \Rightarrow \text{handle } h_{\mathcal{C}} \tilde{E}[(\lambda y. \tilde{e}_{5a}[x := \tilde{v}_2]) \tilde{e}_7]$ by β , where $e_7 = \text{mon}_j^{l,j}(\kappa_1, v_2)$

The target checks the indy argument $\text{mon}_j^{l,j}(\kappa_1, v_2)$ first, the source checks the argument $\text{mon}_j^{l,k}(\kappa_1, v_2)$ first. For $\kappa_1 = \text{flat}(e_p)$ both become $\text{check}_j^l(e_p v_2, v_2)$, so we pair the two checks.

- Case the check fails.

Both programs reach $\text{check}_j^l(\text{false}, v_2)$, then blame_j^l and $\text{handle h}_{\mathcal{C}} \text{error}_l$ as in the case $\text{check}_j^k(\text{false}, v)$.

- Case the check succeeds, $\text{mon}_j^{l,j}(\kappa_1, v_2) \rightarrow^* v_y$ ($v_y = v_2$ for flat κ_1 , otherwise $v_y = \text{guard}_j^{l,j}(\kappa_1, v_2)$).

Let $e_6 = \text{mon}_j^{k,l}(\kappa_2, v_1 \text{mon}_j^{l,k}(\kappa_1, v_2))$, then $e_2 = E[\{\text{mon}_j^{l,j}(\kappa_1, v_2)/y\}e_6]$.

$$e'_1 \Rightarrow^* \text{handle h}_{\mathcal{C}} \tilde{E}[\tilde{e}_6[y := \tilde{v}_y]] = e'_2$$

By purity, the target's argument check $\text{mon}_j^{l,k}(\kappa_1, \tilde{v}_2)$ succeeds again. Each later evaluation of $\text{mon}_j^{l,j}(\kappa_1, v_2)$ inside κ_2 in the source gives v_y again and is matched by zero target steps.

$$\{\text{mon}_j^{l,j}(\kappa_1, v_2)/y\}e_6 \sim \tilde{e}_6[y := \tilde{v}_y] \text{ by the indy row.}$$

Obtain $\text{EQ}(e_2, e'_2)$ (case 1).

- Case e_3 is a base reduction ($\beta, \mu, \text{injl}, \text{injr}, \text{ife}, o$)

$\tilde{e}_3$ is the same redex in **Effects**, and $\tilde{e}_3 \Rightarrow \tilde{e}_4$ by the same rule, using Lemma 5.2 for β and μ .

$$e'_1 \Rightarrow \text{handle h}_{\mathcal{C}} \tilde{E}[\tilde{e}_4] = e'_2$$

Obtain $\text{EQ}(e_2, e'_2)$ (case 1) by Lemma 5.1.

□